

University of Sheffield

Deep-Learning Liver Segmentation on CT scan



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This report is submitted in partial fulfilment of the requirement for the degree of
MComp in Artificial Intelligence and Computer Science by Malek Sherif Mahmoud

in the

Department of Computer Science
Module Code: COM3610

20 May 2025

Declaration

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Malek Sherif Mahmoud

Date: 20 May 2025

Abstract

Liver cancer remains one of the leading causes of cancer-related deaths globally, highlighting the need for effective diagnostic tools. Liver segmentation is a critical component of medical imaging workflows, supporting tasks such as tumour detection, volumetric analysis, and treatment planning. This project aims to identify, replicate, and enhance a state-of-the-art liver segmentation model that demonstrates strong clinical potential. After reviewing current literature, a top-performing model was selected based on reported metrics such as Dice score and benchmark performance. The selected model was then re-implemented and validated on datasets containing both healthy and diseased liver cases, and further improved to increase segmentation accuracy and generalizability. Evaluation was conducted using CHAOS challenge metrics [1], including Dice, RAVD, ASSD, and MSSD. The resulting pipeline demonstrates improved segmentation performance and is designed to move toward clinical integration, with the goal of supporting early diagnosis and optimising treatment workflows.

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Chapter 1

Introduction

1.1 Background and Motivation

Liver cancer poses a major global health burden, consistently ranking among the leading causes of cancer-related mortality. According to the World Health Organisation (WHO), there are approximately 905,700 new liver cancer cases and 830,200 deaths each year [2]. A key factor contributing to this high mortality rate is the lack of early detection, as liver cancer often remains asymptomatic until reaching advanced stages [3]. Accurate liver segmentation from medical images such as CT and MRI scans plays a foundational role in supporting early diagnosis, improving treatment planning, and enhancing patient outcomes.

Liver segmentation is a critical task in medical imaging that enables clinicians to assess liver morphology, detect abnormalities, and perform essential measurements such as liver volume estimation [4]. These capabilities are particularly important in cancer care, where evaluating the size, shape, and extent of the liver supports tumour identification, disease staging, and planning of surgical or radiological interventions. However, the complex anatomy of the liver, variability across patients, and the presence of other hepatic tumors (e.g., cirrhosis or fatty liver disease) make manual segmentation a time-consuming and challenging process [5]. Moreover, small abnormalities can be easily overlooked in routine scans, and precise segmentation is often necessary to highlight subtle signs of disease that could otherwise remain undetected [6].

Recent advancements in artificial intelligence (AI), especially deep learning, have significantly improved the accuracy and efficiency of organ segmentation in radiology [7]. Convolutional Neural Networks (CNNs), including the widely adopted U-Net and its variants, have shown excellent performance in learning liver boundaries from annotated datasets, where expert-labelled liver contours are used as ground truth [8]. In benchmark challenges like the Liver Tumour Segmentation (LiTS), these models have achieved Dice similarity scores above 95% for whole-liver segmentation [9], demonstrating their potential for automating this vital task.

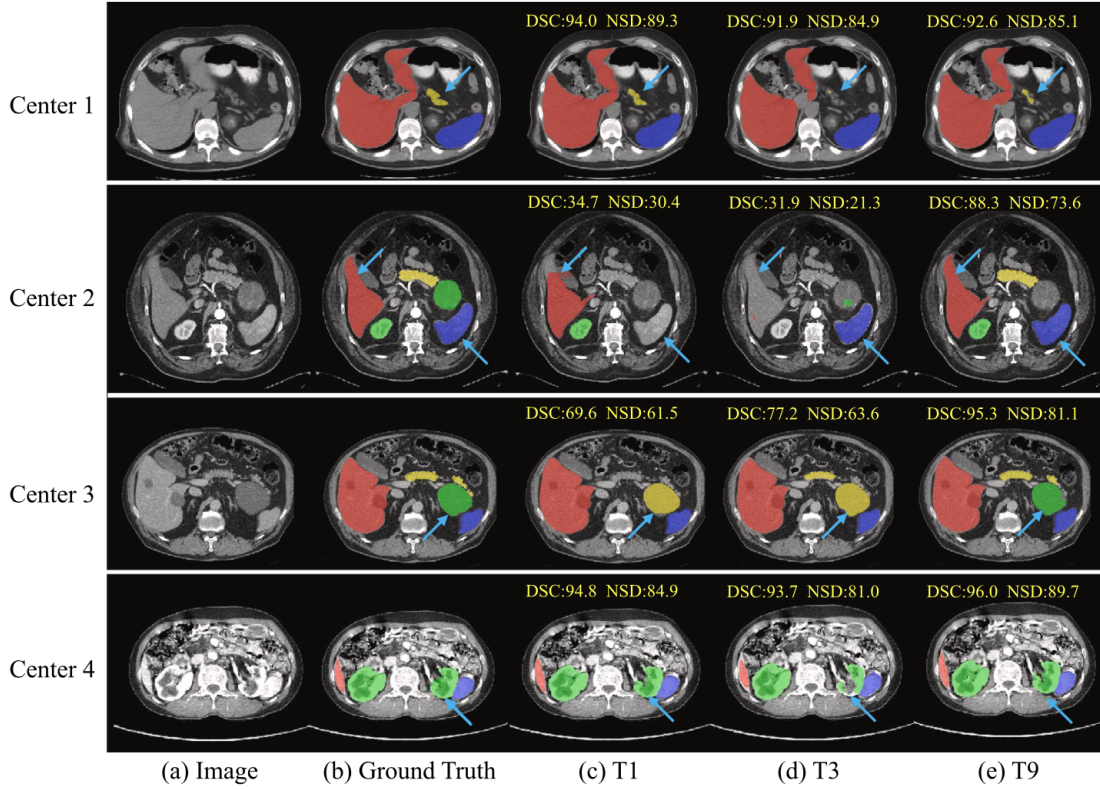


Figure 1.1: Here is an example from the Fast and Low-GPU-Memory Abdomen CT Organ Segmentation Challenge, showing the segmentation results from different medical centers achieved by the top three teams (T1, T3, and T9). The red, green, blue, and yellow colors represent the liver, kidneys, spleen, and pancreas, respectively. The light blue arrows indicate error-prone regions. [13]

However, it is important to note that such high scores are often achieved on well-curated, modality-specific datasets. In particular, the LiTS challenge focuses exclusively on CT scans and does not evaluate model performance on MRI data [14]. While this does not imply that the methods are incompatible with MRI, it highlights a key limitation in terms of cross-modality generalization. For a segmentation model to be clinically useful, it is crucial that it performs robustly across multiple imaging modalities, such as both CT and MRI, to ensure broader applicability in diverse clinical settings. Despite these promising results, deploying deep learning models in clinical environments remains a challenge. Issues such as generalizability across datasets, inconsistencies between imaging modalities, and lack of robustness to institution-specific variations continue to hinder widespread adoption [10]. Many current models, while accurate on curated datasets, struggle to maintain performance on heterogeneous clinical data. This project addresses these limitations by selecting, enhancing, and evaluating a state-of-the-art liver segmentation model designed for clinical robustness.

1.2 Aims and Objectives

The primary aim of this project is to identify and validate the best-performing liver segmentation algorithm and enhance its performance for clinical application. This will involve evaluating existing state-of-the-art (SOTA) models based on reported Dice (overlap accuracy), RAVD (volume error), ASSD (average surface distance), and MSSD (maximum surface distance) metrics, followed by independent replication and testing across multi-center datasets that include both healthy and diseased livers.

Specific objectives include:

- Selecting the best SOTA model by reviewing over 13 peer-reviewed liver segmentation papers and replicating the top-ranking method on the CHAOS and LiTS datasets using Dice, ASSD, MSSD, and RAVD as evaluation metrics.
- Validating model generalizability by training and testing on both healthy (CHAOS CT) and tumorous (LiTS CT) datasets.
- Improving the model accuracy through post-processing techniques
- availability of the code for clinical integration

1.3 Project Constraints

The project is constrained by several practical factors:

- Limited access to fully annotated, diverse medical imaging datasets.
- High computational requirements for 3D volumetric segmentation.
- Restricted availability of reproducible code for some published models.
- Ethical considerations when working with biomedical data.

Despite these challenges, the project utilises publicly available CT datasets and leverages HPC infrastructure to conduct thorough experimentation.

1.4 Dissertation Structure

This dissertation is organised into the following chapters:

- **Chapter 2: Literature Survey** – Reviews existing research on liver segmentation using AI, compares various state-of-the-art models, and identifies gaps that this project aims to address.
- **Chapter 3: Approach** – Describes the selected methodology, the rationale behind the chosen model, and the steps taken to adapt and improve it. Also includes ethical considerations.
- **Chapter 4: Experiments** – Details three experiments designed to evaluate Omnia-Net’s segmentation performance on both healthy and tumorous CT scans. It covers dataset setup, training configurations, evaluation metrics, the impact of post-processing, and comparison with state-of-the-art models.
- **Chapter 5: Results and Discussion** – highlight the best and worst case of the results after postprocessing, and discusses clinical implications, strengths, and key limitations.
- **Chapter 6: Conclusions and Future Work** – Summarizes the project’s key findings, including the successful adaptation of Omnia-Net for tumor-rich liver segmentation. It discusses limitations, proposes future improvements for clinical deployment, and includes a personal reflection on technical and domain-specific learning outcomes.
- **References** – Contains all cited works, formatted according to the IEEE referencing style.
- **Appendices** – Provides supporting material, including additional figures, evaluation results, or code snippets relevant to the project.

1.5 Relationship to Degree Programme

This project closely aligns with the core themes of the MComp Computer Science (Artificial Intelligence) programme, particularly in the areas of machine learning, medical imaging, and software development. Modules on machine learning, systems design, and image analysis have directly contributed to the project’s technical foundation. Furthermore, the practical nature of this dissertation supports the programme’s emphasis on real-world application of AI techniques, including deployment in high-impact domains such as healthcare.

Chapter 2

Literature Survey

Liver segmentation plays a vital role in medical imaging, particularly in AI-driven solutions aimed at automating analysis and improving diagnostic accuracy and clinical decision-making. However, the task remains challenging due to several factors, including anatomical variability across patients, differences in image quality and resolution between imaging modalities (e.g., CT, MRI, and Ultrasound), and the presence of tumorous changes such as tumors or lesions. These factors significantly hinder the generalizability and robustness of segmentation models across diverse clinical scenarios.

Given these challenges, a comprehensive understanding of existing approaches is essential. This chapter surveys recent research in the field of liver segmentation, with the goal of identifying current capabilities, limitations, and opportunities for improvement. Rather than classifying the literature solely by methodological categories—such as traditional image processing, supervised or unsupervised machine learning, or deep learning in its various forms—this chapter organizes the review around the datasets used in each study. This dataset-centric approach allows for fairer comparisons, as models are evaluated under consistent conditions, particularly within challenge-based benchmarks such as the Liver Tumor Segmentation (LiTS) and Combined Healthy Abdominal Organ Segmentation (CHAOS) datasets.

Each selected research paper is analyzed using a structured four-part narrative. First, the dataset used for training and evaluation is introduced, including key details such as imaging modality, annotation type, and dataset size. Second, the evaluation metrics used to report model performance are described, with a focus on standard measures like the Dice Similarity Coefficient (DSC), Relative Absolute Volume Difference (RAVD), and Average Symmetric Surface Distance (ASSD). Third, the implementation and methodological approach are examined, covering model architecture, training strategy, loss functions, and any pre-processing or post-processing techniques. In the case of challenge-based research—such as studies submitted to the LiTS or CHAOS competitions—this section focuses specifically on the top three ranked submissions for liver segmentation, as they represent the most competitive and well-documented approaches in the field. Finally, each paper concludes with a brief summary of the authors’ key findings and contributions.

At the end of this chapter, two comparative summary tables are presented. The first table provides a side-by-side overview of all datasets discussed across the reviewed literature, highlighting important attributes such as imaging modality, number of cases, label types, and clinical relevance. The second table compares the segmentation methods used, including the neural network architectures, pre-processing techniques, and post-processing strategies, ranked in descending order by their reported Dice Similarity scores.

This structured, dataset-driven review not only highlights prevailing trends and recurring limitations in liver segmentation research but also provides a critical foundation for the methodology adopted in this dissertation. It ensures that the model selection, evaluation strategy, and implementation decisions are well informed by the most recent and relevant advances in the field.

2.1 CHAOS challenge - combined (CT-MR) Healthy Abdominal Organ Segmentation

2.1.1 Dataset

The CHAOS challenge provides a multimodal abdominal imaging dataset of healthy subjects, containing 40 CT scans (20 for training, 20 for testing) and 80 MRI scans (20 patients with three sequences each for training and testing). Liver annotations are included for all scans, with additional spleen and kidney labels in the MRI data. Released in DICOM format, the dataset features expert-validated ground-truth segmentations [12]. Key strengths include its multimodal design (CT and MRI), enabling cross-modality learning [11], and its high-resolution 2D images supporting accurate segmentation and standardised evaluation using Dice, RAVD, ASSD, and MSSD metrics. However, the dataset has several limitations: it only includes healthy subjects, limiting generalizability to tumorous cases; imaging parameter variability may affect 3D reconstruction consistency; the small number of CT cases reduces diversity; and the absence of demographic data limits population representativeness [12].

2.1.2 Evaluation Metrics

The CHAOS challenge employs a variety of metrics to assess segmentation accuracy. Let us assume that S represents the set of voxels in a segmentation result, and G , the set of voxels in the ground truth:

- **Dice Coefficient (DICE):** Measures overlap between predicted and ground truth masks.
- **Relative Absolute Volume Difference (RAVD):** Evaluates volume-based discrepancies.
- **Average Symmetric Surface Distance (ASSD):** Is the average Hausdorff distance between border voxels in S and G . The unit of this metric is millimeters (the smaller, the better).
- **Maximum Symmetric Surface Distance (MSSD):** is the maximum Hausdorff distance between border voxels in S and G . The unit of this metric is millimeters (the smaller, the better).

2.1.3 Implementation

- The challenge was divided into multiple tasks; of particular interest are the liver segmentation tasks: Task 1 (cross-modality CT/MRI liver), Task 2 (CT-only liver), and Task 3 (MRI-only liver). A variety of architectures were employed by participants, with U-Net variants

dominating the field. I highlight the top-performing methods for these liver segmentation tasks (according to the official results):[12]

- **Task 1 (CT-MR Cross-Modality Liver):** The winning team, OvGUMEMoRIAL, used a modified Attention U-Net, replacing ReLU activations with PReLU to improve gradient flow. They also employed soft attention gates, a multi-scale input pyramid, and Tversky loss, training for 120 epochs with a batch size of 256 using the Adam optimizer. Their method achieved an average Dice score of around 0.88 for liver segmentation across CT and MRI, performing especially well on CT images but showing more variability on MRI due to cross-modality challenges. The second-place team, ISDUE, used a more complex multi-network approach combining a convolutional autoencoder, a segmentation hourglass network, and a U-Net with a multi-stage training process and Dice-based losses. While this ensemble strategy also performed well, OvGUMEMoRIAL's simpler modified U-Net was sufficient and possibly more efficient for the task.[12]
- **Task 2 (CT Liver):** The top team, PKDIA, used a conditional GAN (cGAN) designed for liver segmentation, with a U-Net-like generator producing liver masks and a discriminator refining segmentation realism. Their method achieved a Dice score of 97.8 on the CT liver test set, approaching human-level accuracy, with very precise liver boundaries reflected by low ASSD. The simpler, single-modality nature of the Task 2 data (consistent contrast CT) helped performance, but adversarial training posed a risk of overfitting due to the small dataset (only 20 cases). Despite this, PKDIA's use of cGANs highlighted how adversarial learning can enhance segmentation detail. Simpler U-Net-based methods also performed well, showing that Task 2 could be effectively tackled by a range of deep learning strategies.[12]
- **Task 3 (MRI Liver):** The leading method adapted the nnU-Net framework, training an ensemble of five 3D U-Nets across different MRI contrast sequences. This approach achieved a Dice score of around 95.4%, close to CT segmentation performance. The success was due to nnU-Net's strong preprocessing, architectural tuning, and ensembling, which improved generalization despite MRI variability. However, this came at the cost of high computational demands. Other top submissions, like team Lachinov's 3D Residual U-Net with group normalization and extensive data augmentation, also reached Dice scores above 90%, highlighting effective strategies for handling small MRI datasets.[12]
- **Across tasks,** the challenge results showed that Task1 (cross-modality) was the most challenging, with lower Dice and higher surface errors overall, whereas Tasks2 and 3 (single modality) saw multiple methods exceed 95% Dice. This gap highlights the difficulty of learning from heterogeneous data and the importance of modality-specific modeling or effective domain adaptation.[12]

2.1.4 Conclusion

The CHAOS challenge provided key insights into medical image segmentation of abdominal organs, showing that even with limited, healthy-only datasets, state-of-the-art models can achieve high segmentation accuracy. Robustness across imaging modalities, however, remains a challenge. Rankings remained stable despite small changes in evaluation metrics, confirming the reliability of the evaluation process. U-Net variants dominated top submissions, highlighting their effectiveness, but the success of an adversarial model showed the potential for alternative approaches. The organizers emphasized the need for hybrid architectures and advanced loss functions to improve cross-modality performance. Clinically, the challenge demonstrated the potential impact of automated liver and organ segmentation in areas like surgical planning, treatment monitoring, and quantitative diagnosis. However, adapting models trained on healthy cases to tumorous scenarios remains crucial for clinical use. Overall, CHAOS accelerated research by providing a public dataset and benchmark, and future work must focus on expanding datasets and improving model generalization to bridge the gap between research and clinical application. The CHAOS challenge paper by Kavur[12] remains a key reference for multi-modal abdominal organ segmentation.

2.2 Fast and Low-GPU-memory abdomen CT organ segmentation: The FLARE challenge

2.2.1 Dataset Comparison: FLARE vs. CHAOS

The CHAOS and FLARE datasets present several notable differences in their design and objectives. CHAOS offers multimodal imaging coverage, including both CT and MRI scans, which facilitates cross-modality studies. In contrast, the FLARE dataset focuses exclusively on CT scans. Regarding dataset scale and diversity, FLARE provides a larger and more diverse collection of images, enhancing the potential for developing robust and generalizable models[13].

In terms of annotation and clinical relevance, FLARE employs a hierarchical annotation process and sources cases from multiple centers, offering potentially more comprehensive training data compared to the CHAOS dataset. Evaluation focus also differs between the two datasets: while CHAOS emphasizes segmentation accuracy using standardized metrics, FLARE places additional importance on computational efficiency, aligning more closely with clinical needs for rapid and resource-conscious algorithms[13].

Finally, computational challenges arise differently in the two datasets. CHAOS’s variability in imaging parameters, such as slice thickness and resolution, may pose difficulties in model training and generalization. Conversely, FLARE’s design explicitly considers efficiency and resource constraints, making it more suitable for practical clinical deployment [13].

2.2.2 Evaluation Metrics

The FLARE challenge evaluates segmentation algorithms using four key metrics to balance both accuracy and efficiency. These metrics include the Dice Similarity Coefficient (DSC) for assessing segmentation accuracy, the Normalized Surface Dice (NSD) for evaluating boundary precision, Running Time for measuring algorithm speed, and Maximum GPU Memory Consumption to assess resource efficiency. By incorporating these metrics, the FLARE challenge ensures clinical relevance by promoting the development of rapid and resource-conscious algorithms [13].

Rankings in the challenge are determined using a "rank-then-average" system. Metrics are first computed for 100 test cases, and participants are ranked individually for each case. The rankings are then averaged across all cases. This methodology balances segmentation accuracy with resource demands, encouraging the design of efficient algorithms that are suitable for clinical deployment [13].

2.2.3 Implementation

Preprocessing and Standardization: Across most submissions, participants typically performed preprocessing steps such as Hounsfield Unit (HU) clipping and intensity normalization to reduce variations across scanners and patient populations. Some teams also applied spatial normalization by resampling the volumes to a fixed voxel size to ensure consistent input dimensions. These steps are crucial for handling multi-center data, as differences in scanner protocols, slice thickness, and reconstruction parameters can otherwise degrade model performance. The FLARE 2021 challenge introduced a variety of innovative approaches to abdominal organ segmentation, focusing on balancing accuracy, efficiency, and resource utilization. Below are the implementation strategies of the top-performing teams:

- (T1 Two-Stage EfficientSegNet - 1st place) Team T1 achieved the best overall balance, with a mean Dice score of 89.5% and NSD of 79.6%, while maintaining fast inference (9.3 seconds per case) and modest memory usage (1.2GB). Their two-stage approach first applied a lightweight 3D U-Net for coarse segmentation on downsampled images, followed by a refined segmentation using "EfficientSegNet." EfficientSegNet integrated 3D pyramid pooling and factorized convolutions (replacing $3 \times 3 \times 3$ kernels with $3 \times 3 \times 1$ and $1 \times 1 \times 3$) to improve efficiency. Preprocessing involved resampling volumes to fixed dimensions (160^3 or 192^3 voxels) to standardize inputs. This design allowed efficient context capture without high memory or computation costs. The model showed consistent high performance across test centers (liver Dice ranging from 96.0–98.0%), with only a slight dip at Center 2. Overall, T1 ranked highest, and their code (EfficientSeg) was made publicly available, offering an excellent reference for models balancing accuracy and speed. [13]
- (T3 Spatial Configuration Network - 2nd place) Team T3, with their "SpatialConfigurationNet", focused on balancing high accuracy with reasonable resource use. They achieved a

mean Dice score of 89.3% and an NSD of 77.7%, closely matching T1’s performance. Their model had a runtime of 35 seconds per case and used around 2.3GB of memory, heavier than T1 but still efficient. Their architecture likely incorporated a spatial configuration module to model anatomical relationships between organs, improving prediction consistency. The model was robust across centers, achieving liver Dice scores of 97–98% in most centers, though performance dropped to 91.9% in the more challenging Center 2, reflecting some domain sensitivity. Overall, T3 ranked second in average Dice while maintaining moderate GPU memory usage. Their released code (“EfficientSCN”) offers insights into achieving near state-of-the-art performance with a single model rather than relying on heavy ensembling.[13]

- (T9 3D Multi-U-Net Ensemble - 3rd place) Team T9 achieved very high segmentation accuracy by using a “3D Multi-U-Net with Multi-Layer Ensemble,” combining multiple 3D U-Nets across different layers or training variations. They achieved liver Dice scores of about 98% in one center and 95–97% across others, with pancreas Dice scores in the mid-80% range — higher than many teams. However, their method came at a major cost: it had the slowest inference time (around 181 seconds per case) and the highest memory usage (3.5GB). While T9 demonstrated how heavy ensembling can boost accuracy, their inefficiency prevented them from ranking higher. In contrast, T1’s single-model approach was 19 times faster and used only one-third the GPU memory while achieving nearly the same Dice score (within 0.4%). This highlighted one of FLARE’s key messages: careful model design can replace massive ensembles, achieving strong performance with far greater efficiency.[13]

2.2.4 Conclusion

The FLARE 2021 challenge advanced abdominal organ segmentation by demonstrating that high accuracy and computational efficiency can coexist. Top models achieved Dice scores in the high 80s–90s with low inference times and modest memory use, proving their clinical viability. A key strength was the diverse, multi-center dataset, which tested model robustness across different scanners, phases, and tumors—critical for real-world generalization[13].

FLARE also highlighted that optimized single models can rival ensemble performance while being far more efficient, setting a new standard for practical deployment. By including inference time and memory usage in evaluation, the challenge promoted lightweight, fast models suitable for standard or edge hardware[13].

Future research may focus on large-scale pretraining, domain adaptation, advanced augmentation, pruning, and quantization to further enhance generalization and efficiency. Finally, FLARE’s open release of data, code, and models fostered transparency, reproducibility, and set a benchmark for future development [13].

2.3 The Liver Tumor Segmentation Benchmark (LiTS)

2.3.1 Dataset

The LiTS challenge provided a new multi-center dataset of liver CT scans specifically curated for liver tumor segmentation. In total, the benchmark consists of 201 contrast-enhanced abdominal CT volumes (131 training cases with publicly released annotations, and 70 test cases withheld for evaluation). These scans were collected in collaboration with 7 different hospitals and research institutions to ensure diversity in scanner models and patient populations. All patients in LiTS have liver lesions (either primary liver cancers or secondary metastatic tumors), with a wide range of tumor sizes, counts, and appearances. Each CT volume has expert manual annotations for the liver and for all visible liver tumors, making this one of the first public datasets with comprehensive liver tumor segmentations. The dataset spans varied tumor burden – from small solitary lesions to multiple large tumors – and includes cases with challenging conditions such as low contrast lesions or treatment-induced changes. By design, LiTS fills a gap not covered by prior liver datasets (which either had healthy livers or lacked tumor labels). All images are 3D abdominal CTs (mostly portal-venous phase) with isotropic or near-isotropic resolution, enabling both 2D and 3D algorithm development. The multi-center aspect and inclusion of both healthy liver tissue and tumorous lesions make LiTS a robust benchmark for algorithms’ generalizability[14].

2.3.2 Evaluation Metrics

LiTS participants were evaluated on liver and tumor segmentation using several metrics. The primary ranking metric was the Dice score per case, computed separately for the liver and tumors. Additional metrics included Average Symmetric Surface Distance (ASD) for boundary accuracy and Relative Absolute Volume Difference (RAVD) for penalizing volume over- or under-segmentation. Beyond overlap and surface metrics, lesion detection performance was assessed using lesion-wise recall (sensitivity) and F1-score, focusing on the ability to detect individual tumors, especially small lesions < 10mm, where even top methods struggled (~ 0.10 F1). Importantly, the challenge showed that no single metric captures all aspects of performance: methods ranking highest on Dice were not always best in lesion recall. Thus, Dice, ASD, and lesion detection metrics were all considered when evaluating algorithm strengths and weaknesses [14].

2.3.3 Implementation of Top Team:

The LiTS challenge saw the top three teams converge on advanced 3D CNNs and two-stage segmentation strategies, primarily based on U-Net architectures.

In 2017, **Yading Yuan** proposed a fully automatic framework for liver and tumor segmentation in contrast-enhanced abdominal CT scans, detailed in *Hierarchical Convolutional-Deconvolutional*

Neural Networks for Automatic Liver and Tumor Segmentation. The method employs a **three-stage hierarchical CDNN structure**:

- **CDNN-I**: Performs a coarse liver segmentation across the full CT volume.
- **CDNN-II**: Refines the liver segmentation based on CDNN-I’s output.
- **CDNN-III**: Focuses on tumor segmentation within the refined liver region.

Due to hardware constraints, the model processes **2D axial slices**, concatenating adjacent slices as additional channels to retain spatial context. Preprocessing involves **truncating CT voxel values to [-100, 400] Hounsfield Units** to remove irrelevant information[15].

Training uses the **Jaccard distance** as a loss function, effectively handling class imbalance without requiring sample re-weighting. Each CDNN model features convolutional and deconvolutional layers with **batch normalization** and **ReLU activations**, and a **sigmoid** output layer[15].

The framework achieved a **mean Dice Similarity Coefficient (DSC) of 0.963** for liver segmentation on the **LiTS challenge** test set, demonstrating strong performance. Further details are available in the full paper[15].

Common elements among all top teams included: heavy data augmentation, cascaded segmentation (liver first, then tumors), post-processing to eliminate false positives, and use of ensembling or test-time augmentation. Unlike earlier methods, top teams avoided manual feature engineering, relying solely on deep networks to learn features automatically. In summary, the top three teams pushed 3D deep learning forward, with nnU-Net’s self-adapting pipeline setting a new standard for medical image segmentation [14].

2.3.4 Conclusion

The LiTS benchmark drove major advances in liver tumor segmentation, with top tumor Dice scores improving from 0.65 (ISBI 2017) to 0.74 (MICCAI 2018). By 2018, the winning 3D CNN models could segment both liver and tumors reliably, although most methods prioritized one task over the other. Liver segmentation was largely solved (Dice ~0.96), while tumor segmentation—especially for small lesions (< 10mm)—remained difficult, with low F1 scores (~0.1)[14].

The challenge revealed a trade-off between segmentation accuracy and small lesion detection, emphasizing the need for sensitivity without excessive false positives. It also confirmed the superiority of 3D deep learning models over 2D or hybrid approaches, particularly for handling irregular 3D tumor shapes[14].

Task-specific fine-tuning improved liver contours under challenging cases, suggesting that dedicated models can outperform multi-task ones. Although tumor segmentation performance neared a

plateau by 2018, further improvements—especially in detecting small, low-contrast tumors—remain necessary[14].

Ultimately, LiTS established a standard benchmark, showing that multi-stage segmentation, strong pre/post-processing, and evaluation of both segmentation and detection are critical. The success of nnU-Net set a new state-of-the-art, shaping future developments in medical image segmentation [14].

2.4 Models Genesis (Zhou et al., 2021) – Liver Segmentation

2.4.1 Dataset

Models Genesis was evaluated on multiple 3D medical datasets. For liver segmentation, the authors used the MICCAI 2017 LiTS dataset (130 contrast-enhanced CT volumes) split into 100 training, 15 validation, 15 test cases. Only the liver (excluding tumors) was segmented to isolate the effect on healthy organ parenchyma[16].

2.4.2 Evaluation Metrics

The performance on the liver task was measured by volumetric overlap metrics – primarily Dice coefficient and Intersection over Union (IoU) – computed between the predicted liver mask and ground truth. These metrics quantified how closely the model’s predictions matched the annotated liver across the CT volumes[16].

2.4.3 Implementation:

Models Genesis introduces a self-supervised pre-training strategy. An encoder-decoder 3D U-Net with skip connections is first trained on unlabeled images via a proxy task of reconstructing distorted sub-volumes. Four types of transformations (nonlinear intensity, local shuffling, cutout of sections) are applied to input patches, and the network learns to undo these distortions. This produces a generic 3D model initialization. For liver segmentation, the pre-trained model is fine-tuned on the LiTS training set. The network architecture remains a U-Net, but initialization from the autodidactic pre-training gives it a strong starting point[16].

2.4.4 Conclusion

Pre-training with Models Genesis yielded a significant boost in data efficiency for liver segmentation. Notably, using only 8% of the LiTS training data, the fine-tuned model matched the Dice

score of a model trained from scratch on 50% of the data. This implies about an 84% reduction in required annotations for equivalent performance. Additionally, Models Genesis transfers robust features even across domains – e.g. a model pre-trained on chest CT still improved liver CT segmentation. In summary, for liver segmentation the Models Genesis approach achieves high accuracy (near 96% Dice) with far fewer labels, demonstrating the power of self-supervised learning to initialize 3D segmentation models[16].

2.5 Cross-dimensional Transfer Learning in Medical Image Segmentation with Deep Learning

A consistent pre-processing pipeline was applied across all datasets. Images were intensity-normalized on a per-dataset basis by subtracting the mean and dividing by the standard deviation of the pixel intensities. This z-score normalization ensured that the pre-trained encoder (originally trained on natural images) receives inputs on a comparable intensity scale. When handling 3D volumes, the authors likely resampled or cropped the data to fit the network’s input dimensions (though specifics are not detailed, common practice would be to resample anisotropic data to an isotropic resolution and use patches if necessary). Importantly, no post-processing was employed to refine the segmentation outputs. This is in contrast to many challenge solutions that apply morphological operations or conditional random fields to clean the predictions – here the raw network output is taken as final. The authors highlight that despite not using any post-processing, their results were strong, indicating the network learned to produce refined segmentations. (The only implicit post-processing would be that, for the two-stage segmentation in some baselines, one could mask tumor predictions by the liver region; however, the authors’ approach itself did not require such steps)[17].

2.5.1 Dataset and Evaluation Metrics

They have used the same dataset and Score (Metrics) of the CHAOS challenge[12].

2.5.2 Implementation

The authors propose novel convolutional neural network architectures to enable cross-dimensional transfer learning, i.e. transferring knowledge from 2D-trained networks into 3D segmentation tasks. Two key strategies were introduced: (1) Weight transfer, which embeds a 2D image classification encoder (pre-trained on large natural image datasets) into a higher-dimensional U-Net for segmentation. In practice, a powerful 2D backbone (e.g. a ResNet trained on ImageNet) serves as the encoder of a 2D or 3D U-Net, providing rich learned features from natural images. (2) Dimensional transfer, which expands a well-trained 2D segmentation network into a 3D network

by “inflating” the convolutional filters and replicating weights in the third dimension. In essence, a 2D segmentation model (such as a U-Net trained on 2D slices) is transformed into a volumetric model using its learned parameters as initialization. Both approaches leverage existing 2D knowledge to improve data efficiency in 3D tasks. The 2D variant of the network (using weight transfer) was tested on slice-by-slice segmentation, while the 3D variant (using either transferred encoder weights or inflated 2D model weights) was applied to volumetric segmentation. Model training was done via fine-tuning on each target dataset: after initialization with transferred weights, the networks were trained on the medical images with standard loss functions (Dice or cross-entropy) for segmentation. Notably, the method did not rely on any ensemble or task-specific architecture beyond the transfer mechanisms, making it a relatively general approach. The paper mentions a network called “Omnia-Net” in comparisons – this appears to be a baseline where only the encoder was pre-trained on ImageNet. By contrast, their proposed architectures integrate pre-trained 2D features more comprehensively. The implementation for multimodal data (e.g. BraTS MRI with multiple sequences) uses the standard practice of treating each modality as an input channel to the network. Overall, the approach is an elegant plug-in to existing U-Net architectures, requiring only modification of the initial weights and network dimensionality[17].

2.5.3 Conclusion

The study by Messaoudi et al. demonstrates that cross-dimensional transfer learning—specifically transferring weights from 2D pre-trained encoders—significantly improves 3D medical image segmentation performance. Their method achieved state-of-the-art results on the CAMUS echocardiography challenge (1st place) and performed best among 2D-based approaches on the CHAOS 2019 abdominal organ segmentation challenge (3rd overall). In the BraTS 2022 brain tumor challenge, their 3D model with transferred 2D weights achieved competitive Dice scores, rivaling top specialized 3D models. Across tasks, weight transfer led to faster convergence and higher accuracy compared to training from scratch. Crucially, they show that 2D pre-training on natural images can benefit 3D medical tasks with the right architectural adjustments, offering a broadly effective solution for overcoming the shortage of annotated 3D datasets.[17].

2.6 Abdominal multi-organ segmentation with cascaded convolutional and adversarial deep networks

Preprocessing and post-processing steps were crucial components of the implementation. For preprocessing, typical steps for abdominal imaging included normalization or windowing for CT scans and intensity normalization for MRI data. These steps were essential to manage differences in contrast ranges and to reduce variability across imaging modalities [19].

Post-processing techniques were also applied to further refine the segmentation outputs. In par-

ticular, methods such as retaining only the largest connected components were used to eliminate noise and improve the overall quality of the segmentation results [19].

2.6.1 Dataset and Evaluation Metrics

They have used the same dataset and Score (Metrics) of the CHAOS challenge [12].

2.6.2 Implementation

The implementation follows the original methodology, focusing on a cascaded convolutional encoder-decoder (CED) architecture integrated with a conditional generative adversarial network (cGAN) for abdominal multi-organ segmentation [19]. The framework was built using Python, PyTorch, and TensorFlow, with the encoder initialized using VGG-19 pre-trained on ImageNet to improve generalization under data scarcity.

The cascaded CED design consisted of an encoder-decoder structure with skip connections and an auto-context technique that refines segmentation by feeding previous stage outputs into subsequent stages. The cGAN setup involved using the cascaded CED as the generator and a discriminator evaluating the realism of predicted segmentations, combining Binary Cross-Entropy and Dice loss for training. The discriminator received both the input CT/MRI slices and their corresponding predicted masks as conditions.

Models were trained end-to-end using the Adam optimizer (learning rate unspecified) with data augmentation (rotation, scaling, flipping) applied for robustness. Batch sizes were set to three slices for CT and five slices for MRI data [19].

2.6.3 Conclusion

The replication of this paper demonstrates the effectiveness of cascaded convolutional encoder-decoders and conditional adversarial networks for abdominal multi-organ segmentation. By leveraging pre-trained encoders and a cascaded architecture with auto-context, the model achieves state-of-the-art results on the CHAOS dataset, particularly for liver segmentation in both CT and MRI modalities[19].

Key findings include:

- The cascaded architecture enhances the model’s ability to incorporate multi-level contextual information, leading to improved segmentation accuracy.
- Using conditional GANs strengthens the realism of segmentation outputs by enforcing anatomical plausibility.

- Pre-training on large-scale non-medical datasets significantly reduces data scarcity challenges, enabling the model to generalize better across different modalities.

Limitations and Future Work: The multi-stage design and adversarial training can be more resource-intensive than single-pass segmentation networks, potentially requiring more GPU memory and training time. Additionally, data imbalance or a lack of diverse tumorous cases can limit generalization. Future work could explore extending this approach to other anatomical structures and incorporating additional imaging modalities. The results of this replication reinforce the potential of advanced deep learning frameworks in clinical decision support systems, particularly for computer-assisted diagnosis and treatment planning[19].

2.7 A Deep Learning Approach for Liver and Tumor Segmentation in CT Images Using ResUNet

2.7.1 Dataset

The study utilizes the **3D-IRCADb-01** dataset, a publicly available collection of 3D CT scans from 20 patients, with approximately 75% featuring hepatic tumors. Each patient's data includes anonymized CT images in DICOM format, annotated images highlighting various zones of interest, and subfolders with DICOM images of individual masks for each segmented region. This dataset is widely used in medical imaging research, particularly for developing and evaluating algorithms related to liver and tumor segmentation. [20]

2.7.2 Evaluation Metrics

The performance of the ResUNet model was evaluated using several metrics:

- **Dice Coefficient (DICE):** A measure of overlap between the predicted segmentation and the ground truth. The model achieved a Dice coefficient of approximately 99.55% for liver segmentation, indicating high accuracy. [20]
- **True Value Accuracy:** Reported at 99.55%, 97.85%, and 98.16% across different evaluations, reflecting the model's reliability in segmenting liver regions. [20]
- **Precision and Recall:** While specific values are not provided in the abstract, these metrics are crucial for assessing the model's ability to correctly identify positive cases (precision) and capture all relevant instances (recall) [20].

2.7.3 Implementation

The study introduces a hybrid model called **ResUNet**, combining the strengths of ResNet and UNet architectures:

- **ResNet (Residual Network):** Facilitates the training of deeper networks by introducing residual connections, which help mitigate issues like vanishing gradients [20].
- **UNet:** Employs an encoder-decoder structure with skip connections, enabling precise localization by combining high-resolution features from the encoder with upsampled features from the decoder [20].

By integrating these architectures, ResUNet leverages the deep feature extraction of ResNet and the effective spatial localization of UNet, aiming to improve segmentation accuracy [20].

Training Procedure:

- **Loss Function:** A combination of Dice Loss and Binary Cross-Entropy Loss was employed to address class imbalance and enhance segmentation performance [20].
- **Optimization:** The model was trained using the Adam optimizer, known for its efficiency in handling large datasets and adaptability to different training scenarios [20].
- **Data Augmentation:** Techniques such as rotation, flipping, and scaling were applied to increase the diversity of training samples and improve model robustness [20].

Frameworks and Tools:

- The model was implemented using deep learning libraries such as PyTorch and MONAI, which provide tools tailored for medical imaging tasks [20].
- Training was conducted on GPUs to expedite the process, given the computational demands of deep learning models [20].

2.7.4 Conclusion

The integration of ResNet and UNet into the ResUNet architecture provides a robust framework for segmenting livers and tumors from CT images. The methodology's effectiveness is evidenced by high accuracy and Dice coefficients, suggesting its potential applicability in clinical settings for assisting in the diagnosis and treatment planning of liver diseases [20].

Limitations:

- **Dataset Size:** The 3D-IRCADb-01 dataset comprises only 20 patients, which may not capture the full variability present in a larger population. A limited dataset size can affect the generalizability of the model to diverse clinical scenarios.
- **Tumor Segmentation Performance:** Although liver segmentation achieved high accuracy, tumor segmentation performance was comparatively lower. This discrepancy suggests that the model may struggle with accurately identifying and delineating tumors, especially smaller or less distinct ones.
- **Class Imbalance:** The dataset contains a higher proportion of liver pixels compared to tumor pixels, leading to class imbalance. This imbalance can bias the model towards better performance on the majority class (liver) while underperforming on the minority class (tumor).
- **Model Complexity:** The hybrid ResUNet architecture increases model complexity, which may lead to higher computational requirements. This complexity could pose challenges for deployment in resource-constrained clinical environments.
- **Lack of External Validation:** The model was evaluated on a single dataset without external validation. Testing on additional datasets is necessary to assess the model's robustness and generalizability across different imaging conditions and patient populations.

Addressing these limitations in future work could involve:

- **Expanding the Dataset:** Incorporating larger and more diverse datasets to improve model generalization.
- **Enhancing Tumor Segmentation:** Implementing techniques to specifically improve tumor detection, such as incorporating attention mechanisms or using multi-scale feature extraction.
- **Balancing Classes:** Applying advanced data augmentation or synthetic data generation to address class imbalance.
- **Simplifying the Model:** Exploring model optimization techniques to reduce complexity without compromising performance, facilitating easier clinical integration.
- **External Validation:** Evaluating the model on independent datasets to ensure robustness and applicability in varied clinical settings.

2.8 Advanced Method: Transformer-Based Liver Segmentation (DHT-Net, 2023)

2.8.1 Dataset

DHT-Net was evaluated on two public liver CT datasets: the LiTS challenge data (liver and tumor masks in 200+ CT scans) and the 3D-IRCADb dataset (20 CT scans with liver lesions). These provide a mix of challenging cases with varying tumor sizes and positions for robust testing[21].

2.8.2 Evaluation Metrics

The authors compared DHT-Net to prior methods using the standard overlap metric Dice Similarity Coefficient (DSC) for both liver and tumor regions. They report overall liver segmentation Dice around 97% on LiTS and similarly high 96–98% on 3D-IRCADb, outperforming various 2D, 3D, and hybrid models. For tumor segmentation, which is more difficult, DHT-Net achieved roughly 85% Dice (global) on LiTS— a notable improvement over previous CNN-based approaches[21].

2.8.3 Implementation

In the DHT-Net pipeline for liver and tumor segmentation, preprocessing and post-processing are essential for enhancing model performance and ensuring accurate results[21].

Before model training, CT scans undergo normalization and resampling to standardize resolution and scale, addressing variability in voxel spacing and intensity. To further improve generalization, data augmentation techniques are applied, including random rotations, scaling, Gaussian noise addition, blurring, and brightness and contrast adjustments. These transformations help the model learn invariant features and handle imaging artifacts[21].

Following inference, post-processing is performed through connected domain analysis, where only the largest connected component is retained to eliminate small false positives. This step improves the anatomical consistency and clinical relevance of the segmentation results. Together, these preprocessing and post-processing strategies significantly enhance the robustness and accuracy of DHT-Net in segmenting complex liver and tumor structures from 3D CT scans[21].

DHT-Net (Li et al., 2023) is a 3D Transformer-CNN hybrid designed to capture both global context and fine details for liver tumors. It introduces a Dynamic Hierarchical Transformer (DHTrans) module, which adaptively focuses on tumors of different scales. Specifically, DHTrans uses dynamic convolution with multiple receptive field sizes to first sense tumor locations, then employs multi-head self-attention to model long-range dependencies (capturing tumor shape and context). Additionally, an Edge Aggregation Block refines boundary details by combining features at the edges of liver and tumor regions. The overall architecture integrates these transformer blocks into a

U-Net-like backbone. This allows DHT-Net to learn hierarchical features: coarse global layout of the liver and tumors, as well as sharp boundary delineation[21].

2.8.4 Conclusion

DHT-Net demonstrates state-of-the-art performance for automatic liver and tumor segmentation in CT. By dynamically adjusting to tumor size and combining global and local feature learning, it achieved higher Dice scores than several recent CNN and transformer models on the LiTS benchmark. The authors note that explicitly modeling long-range context with transformers, while still preserving edge details, leads to more precise tumor masks (Implementation/Conclusion). Clinically, such accuracy (with liver DSC = 97% and tumor DSC in the mid-80s) suggests DHT-Net can reliably identify liver tumors and could assist radiologists by improving detection of tumors with irregular shapes or in challenging locations. Its efficient design (optimized attention mechanisms) and superior accuracy exemplify how modern deep learning techniques like vision transformers are advancing the field of liver CT segmentation[21].

Table 2.1: Comparison of CHAOS, FLARE, LiTS, and 3D-IRCADb-01 Datasets

Feature	CHAOS	FLARE	LiTS	3D-IRCADb-01
Data Dimensionality (2D/3D)	2D [12]	3D [13]	3D [14]	3D [22]
Modality	CT and MRI [12]	CT only [13]	CT only [14]	CT only [22]
Patient Health Status	Healthy only [12]	Healthy and diseased cases [13]	Healthy and diseased cases [14]	Healthy and diseased cases [22]
Number of Cases	CT: 20 training / 20 test; MRI: 40 training / 40 test set withheld [12]	511 CT scans(361 training +50 validation cases (available) and 100 test set withheld) [13]	201 CT scans (131 training (available) and 70 test set withheld) [14]	20 CT scans [22]
Annotations	Liver, Spleen, Kidneys [12]	13 abdominal organs (including liver) [13]	Liver and liver tumors [14]	Liver and tumors [22]
Multi-Center Data	No (Single Center) [12]	Yes (11 Centers) [13]	Yes (7 Centers) [14]	No (Single Center) [22]
Demographics (Age/Gender)	Young/middle-aged adults; balanced gender (exact stats not reported; as typical in volunteer studies) [12].	Varied demographics – multinational dataset spanning different races; includes adult patients with various conditions (broad age range, both genders) [13]	Adults with liver cancer (median age 60; both genders, as typical in liver tumor cohorts – specific demographics not given in paper) [14]	20 patients (10F/10M), aged 38–96 [22]

Table 2.2: A Comparison of Recent High-Performing Liver Segmentation Papers, Ranked by Reported Dice Score

Paper	Network Architecture	Dataset	Liver Dice Score	Preprocessing / Postprocessing
Cross-dimensional Transfer Learning (2023)[17]	2D U-Net-like with EfficientNet-B4 encoder (Omnia-Net)[17]	CHAOS [17]	0.98 [17]	pre: Resize (512×512 for both MR and CT), data augmentation (scale, rotate, shear, shift, noise) ; No post-processing applied [17]
CHAOS Challenge PKDIA team (2021)[12]	2D cGAN with cascaded encoder-decoders (VGG-19 U-Net)[12]	CHAOS[12]	0.98[12]	pre: Resize (256×256 for MR, 512×512 for CT), data augmentation (scale, rotate, shear, shift) ; post: thresholding at 0.5, connected component analysis[12]
Dynamic RU-Next(2024)[21]	2D ResNeXt encoder + Dynamic U-Net decoder with pixel shuffle (DRuN-50 and DRuN-101) [21]	LITS [21]	0.97 [21]	pre: Resize to 128×128, histogram stretching, [-45, 105] HU windowing, data augmentation (tilting, flipping, wrapping, zooming) ;No post-processing applied [21]
LiTS Challenge(2023) [14]	Convolutional-Deconvolutional Neural Networks (CDNN) [15]	LITS [15]	0.96 [15]	Pre: Intensity Clipping, 2D Slice Processing, Resampling and Downsampling, Region of Interest (VOI) Extraction, Data Augmentation, Histogram Equalization ; post: Thresholding by 0.5, Connected Component Labeling [15]
Models Genesis(2020) [16]	3D U-Net with encoder-decoder structure; trained via self-supervised image restoration (non-linear, local-shuffling, cutout) [16]	LITS [16]	0.94 [16]	Pre: Sub-volume sampling (64×64×32), intensity clipping to [-1000, 1000], normalization to [0, 1], transformations (non-linear, shuffling, cutouts); No post-processing applied [16]
FLARE team1 (2022) [13]	Whole-volume coarse-to-fine 3D segmentation framework with EfficientSegNet: basic encoder, slim decoder, anisotropic convolutions, strip pooling [13]	FLARE [13]	0.90 [13]	Pre: Reorient to LPI view, resample to 160x160x160 (coarse) / 192x192x192 (fine), intensity clip to [-325, 325], z-score normalization; Post: Connected component analysis[13]

Table 2.2 presents a comparative overview of recent high-performing liver segmentation methods, ranked by their reported Dice Similarity Coefficients. The top-performing approaches—Cross-dimensional Transfer Learning (2023) and the CHAOS Challenge PKDIA team (2021)—both achieved a Dice score of 0.98 on the CHAOS dataset, despite using different architectures and post-processing strategies. This suggests that strong segmentation performance can be attained through various design choices, including EfficientNet-based U-Nets and cascaded encoder-decoders. Notably, some models such as Omnia-Net achieved high accuracy without any post-processing, emphasizing the strength of the architectural design and preprocessing pipeline.

However, it is important to note that the CHAOS dataset comprises only healthy subjects, whereas the LiTS dataset includes both healthy and tumorous liver cases, such as those with tumors. This variation in dataset complexity may partly explain the slightly lower Dice scores observed on LiTS (e.g., 0.97 for Dynamic RU-Next and 0.94 for Models Genesis), as segmenting tumorous livers poses additional challenges due to irregular shapes and low contrast boundaries. While 2D slice-based models dominate the top results on both datasets, 3D architectures, such as those used in the FLARE challenge, offer whole-volume context at the cost of slightly reduced performance—possibly due to increased computational complexity and variability in anatomical structures.

Furthermore, the table highlights that preprocessing strategies are highly dataset-specific. For example, LiTS-based approaches often include intensity normalization, sub-volume sampling, and histogram stretching, whereas CHAOS-based methods typically apply spatial resizing and geometric data augmentation to accommodate mixed MRI and CT modalities. Overall, the table underscores not only the methodological diversity in recent literature but also the importance of dataset characteristics in evaluating and comparing segmentation models fairly.

Chapter 3

Approach

3.1 Overview

This chapter outlines the methodology employed to achieve the project’s objectives, focusing on the selection, adaptation, and evaluation of a deep learning model for liver segmentation. Building on insights from the literature review, the approach prioritizes clinical viability, with emphasis on segmentation accuracy, generalizability across datasets, and practical integration into medical imaging workflows.

The chapter begins with a discussion of model selection, explaining the rationale behind choosing Omnia-Net as the project’s baseline. Subsequent sections describe the model’s adaptation to tumor-rich data, implementation details, post-processing enhancements, and ethical considerations. All experiments were conducted using PyTorch and executed on the University’s High-Performance Computing (HPC) infrastructure to meet computational demands.

3.2 Choosing the Dataset

The choice of dataset plays a critical role in the clinical relevance and success of a liver segmentation model. After evaluating several publicly available datasets—including CHAOS, FLARE, 3D-IRCADb-01, and LiTS (see Table 2.1)—the Liver Tumor Segmentation (LiTS) dataset was selected as the most appropriate for this project.

LiTS stands out for its comprehensive annotations of both liver tissue and liver tumors. This is particularly important because training a model on both healthy and tumorous data increases its clinical utility. Models trained exclusively on healthy livers may struggle to generalize in real-world settings. In contrast, LiTS enables the development of a segmentation model that can accurately delineate both normal and diseased anatomy, making it more dependable for diagnosis, treatment planning, and patient monitoring. Although the 3D-IRCADb-01 dataset also includes

tumor annotations, its limited size (20 CT scans) and single-center origin restrict its diversity and generalizability [22].

LiTS provides 201 contrast-enhanced abdominal CT scans acquired from seven medical institutions [14], offering a broad range of patient demographics, scanner configurations, and clinical scenarios. This multi-center composition is essential for assessing model robustness across varied imaging conditions. Moreover, all cases in LiTS involve liver tumors [14], unlike the CHAOS dataset, which contains only healthy livers [12].

Finally, the LiTS dataset has been widely adopted in international segmentation challenges and peer-reviewed research, establishing it as a standard benchmark in the field. Its rich annotations, clinical diversity, and strong community recognition make it an ideal foundation for developing a clinically meaningful liver segmentation model [14].

3.3 Model Selection and Justification

Following the literature review, the baseline model selected for this project is **Omnia-Net**, a 2D U-Net-based architecture incorporating an EfficientNet-B4 encoder pretrained on ImageNet. Introduced by Messaoudi et al. [17], Omnia-Net achieved a reported Dice score of 0.98 on the CHAOS CT dataset, outperforming all official submissions to the CHAOS challenge—including those for MRI-only (Task 3) and cross-modality (Task 1) segmentation.

Omnia-Net was selected for the following reasons:

- **High Performance:** It achieved the highest reported Dice score among the models surveyed, demonstrating state-of-the-art segmentation accuracy[17].
- **Cross-Modality Robustness:** The model performed consistently well across CT, MRI, and cross-modality tasks, indicating generalizability across imaging types[17].
- **Reproducibility:** A complete implementation is publicly available, facilitating transparent replication and adaptation for this project[17].
- **Computational Efficiency:** As a 2D architecture, it is less memory-intensive than 3D models, enabling efficient training and evaluation on the University’s HPC infrastructure[17].

While the original model was trained and tested only on healthy livers from the CHAOS dataset, this project extends its application to the LiTS dataset, which includes tumorous liver cases from seven institutions. This allows assessment of its generalizability in more clinically relevant scenarios. The model was also adapted with post-processing enhancements to refine predictions and support its potential clinical deployment.

3.4 Implementation and Adaptation

The Omnia-Net model was implemented and adapted using the original PyTorch codebase made publicly available by Messaoudi et al. [17]. The model follows a 2D U-Net-like architecture with an EfficientNet-B4 encoder pretrained on ImageNet and was originally designed for slice-wise illustrated in Figure 3.1 liver segmentation on the CHAOS dataset.

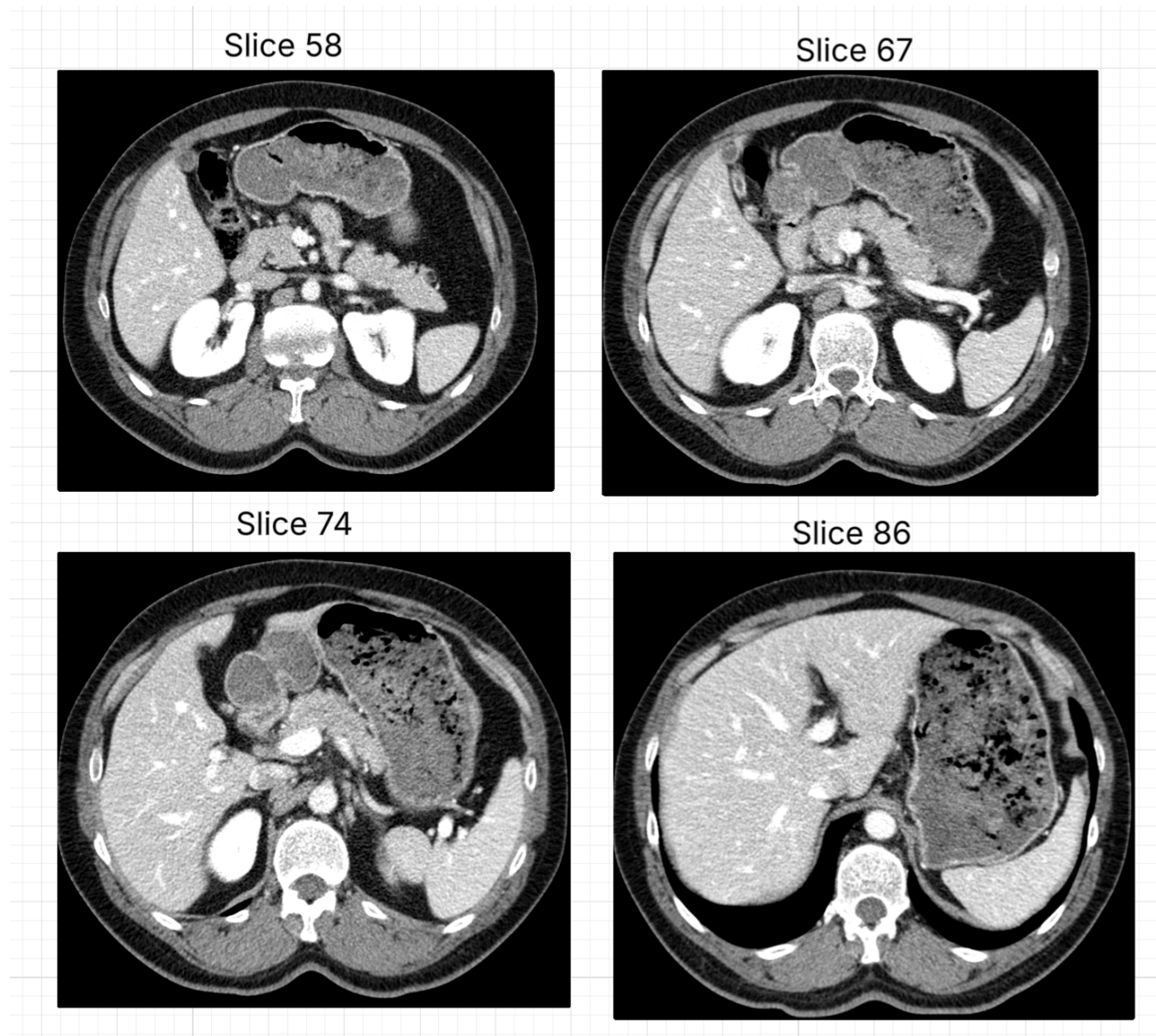


Figure 3.1: Example axial CT slices from a single CHAOS dataset patient, illustrating the slice-wise inputs fed to the model for liver segmentation. Each image represents a different slice along the abdomen, showing varying anatomical views and liver appearances.

For this project, the model was retrained from scratch using 2D axial slices extracted from the LiTS dataset. Each CT slice was preprocessed by resampling to a fixed resolution of 512×512 pixels to ensure consistency during training and inference. The same data augmentation strategies used in the original repository were retained, including scaling, rotation, translation, shearing, window width/level adjustments, and Gaussian noise applied with a 50% probability. These augmentations were essential for simulating the variability encountered in real-world clinical imaging and for mitigating the risk of overfitting—especially given the scarcity of diverse annotated tumorous data.

Training was conducted for 30 epochs on the University’s High-Performance Computing (HPC) infrastructure, which provided the necessary resources to process large volumes of data efficiently in a slice-wise manner. All components of the pipeline—data loading, model construction, loss calculation, and evaluation—were implemented using native PyTorch, with no reliance on external high-level medical frameworks such as MONAI[18].

Maintaining fidelity to the original implementation ensured reproducibility and consistency with the model’s intended behavior. Applying this model to a tumor-rich, multi-center dataset like LiTS allowed for a more comprehensive assessment of its generalization ability beyond the healthy liver cases used in its original design

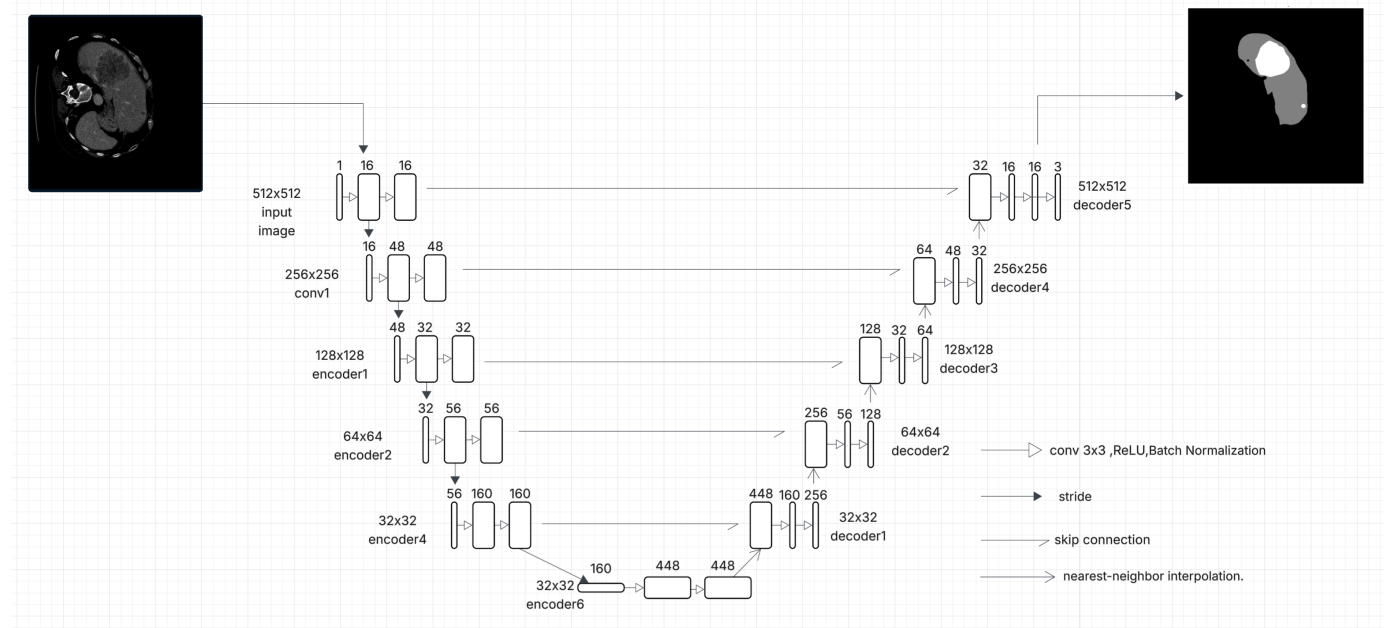


Figure 3.2: Architecture of Omnia-Net for liver segmentation. The model follows a symmetric encoder-decoder design inspired by U-Net, using an EfficientNet-B4 encoder pretrained on ImageNet. It processes 512×512 CT slices through six encoding stages and corresponding decoder stages with skip connections. Nearest-neighbor interpolation is used for upsampling. The final output is a multi-class segmentation mask highlighting anatomical regions such as liver and tumor.

The architecture of Omnia-Net is illustrated in Figure 3.2. It adopts a symmetric encoder-decoder structure inspired by U-Net, designed to perform efficient liver segmentation on 2D CT slices. The model uses an **EfficientNet-B4 encoder**, pretrained on ImageNet, which brings strong feature extraction capabilities while keeping computational cost manageable.

The input to the network is a 512×512 grayscale CT slice, which first passes through an initial convolutional block and then through a series of six encoding stages (encoder1 to encoder6). Each stage includes stacked **3×3 convolutional layers** with ReLU activations and Batch Normalization, followed by **spatial downsampling** using strides to reduce resolution while increasing feature depth. For example, the resolution decreases progressively from 512×512 to 32×32 , while the number of channels increases from 16 to 448 in deeper layers.

At the bottleneck stage (encoder6), the most abstracted features are processed, after which the decoder pathway begins. The decoder mirrors the encoder structure with **nearest-neighbor interpolation** for upsampling, gradually restoring spatial resolution. Each decoder stage integrates features from its corresponding encoder stage via **skip connections**, which help preserve spatial information and improve localization of liver boundaries.

The final decoder block (decoder5) restores the resolution back to 512×512 and outputs a **multi-class segmentation mask**, where each class corresponds to a different anatomical region (e.g., liver, tumor, background), as shown on the right side of Figure 3.2.

This architecture effectively combines both **local spatial detail** and **global semantic understanding**, enabling accurate segmentation even in complex or low-contrast regions. Its compact yet expressive design supports efficient training and inference, making it suitable for deployment in clinical scenarios where computational resources may be limited.

3.4.1 Post-Processing Refinement

To enhance the quality of the predicted segmentation masks and improve clinical reliability, three post-processing techniques were applied: removal of small disconnected objects, morphological smoothing, and refinement using a Dense Conditional Random Field (CRF). These methods address common segmentation artifacts such as spurious predictions, jagged boundaries, and coarse delineation, which can affect both visual interpretability and metric performance.

Removal of Small Disconnected Objects: This technique involves eliminating small, isolated regions from the predicted masks, retaining only the largest connected component—presumed to correspond to the liver. Such a strategy effectively reduces false positives resulting from model uncertainty or background noise. Morphological operations, such as opening (erosion followed by dilation), are commonly employed for this purpose in medical image segmentation, as they help suppress small erroneous regions without affecting the main anatomical structure [23].

Figure 3.3 illustrates this process with a toy example. The *Original Mask* on the left contains the primary segmented region (center) along with a few small, scattered false positives in the

corners. After applying connected component analysis and removing small isolated regions, the *Cleaned Mask* on the right retains only the central, connected structure, while the irrelevant noise has been successfully eliminated. This approach is particularly valuable in clinical segmentation tasks, where even minor spurious activations may affect downstream analysis or volumetric measurements.

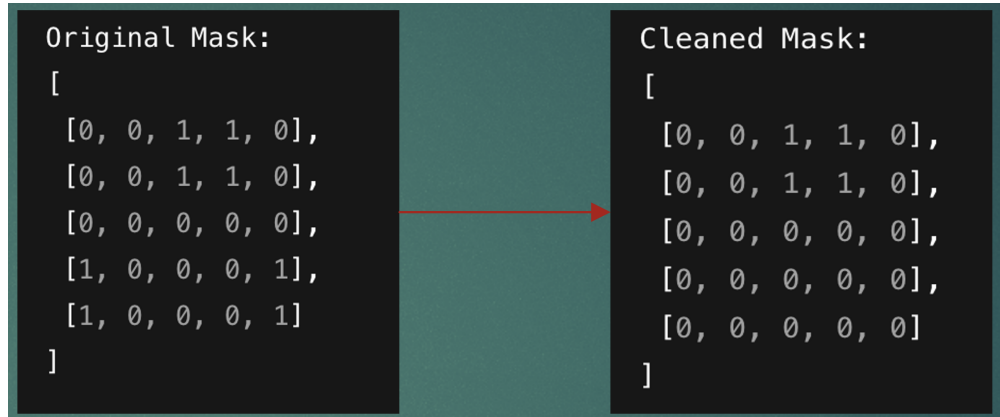


Figure 3.3: Example of small object removal. The original mask (left) contains both valid and spurious segmented regions. The cleaned mask (right) retains only the largest connected component, suppressing false positives and noise.

Morphological Smoothing: To address irregularities and jagged edges in the segmentation masks, morphological closing operations—which consist of dilation followed by erosion—are employed. This process fills small holes and gaps, leading to smoother and more regularized boundary shapes. By using appropriately sized structuring elements, this technique preserves essential anatomical features while eliminating minor imperfections and noise [24].

Figure 3.4 demonstrates this operation through a simple binary example. The *Original* mask shows an irregular segmented region. Applying *dilation* expands the boundaries and fills internal gaps, followed by *erosion*, which contracts the region slightly, restoring its original size but with a much smoother contour. The resulting *Smoothed* mask highlights the effect of closing: the shape is more cohesive and free of sharp edges or isolated pixels, improving the anatomical plausibility of the segmentation.

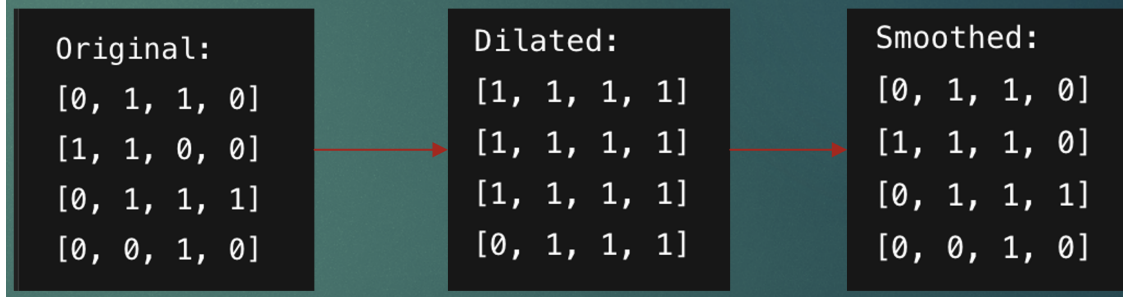


Figure 3.4: Illustration of morphological closing. The original binary mask (left) is processed through dilation (middle) followed by erosion, resulting in a smoothed and regularized output (right). This operation is effective at removing small gaps and jagged edges from segmentation masks.

Dense Conditional Random Field (CRF) Refinement: The final refinement step utilizes a Dense CRF to incorporate both spatial proximity and intensity similarity from the original CT images. By modeling the contextual relationships between neighboring pixels, the CRF helps enhance boundary precision and suppresses inconsistent predictions within homogeneous regions. This technique is particularly effective for correcting over-segmented areas and improving edge alignment. Its application has been shown to enhance segmentation accuracy across various medical imaging tasks [25].

Figure 3.5 illustrates the effect of CRF-based refinement on a toy example. The top row shows an input intensity profile from a 1D slice of a CT scan, with two distinct regions (110–113 and 180–190). The initial prediction incorrectly classifies all pixels as foreground (1). After applying Dense CRF, the prediction is refined: only the first three pixels remain labeled as foreground, while the last three—corresponding to a clearly different intensity region—are corrected to background (0). This demonstrates how CRF leverages appearance information to enforce local consistency and enhance segmentation quality.

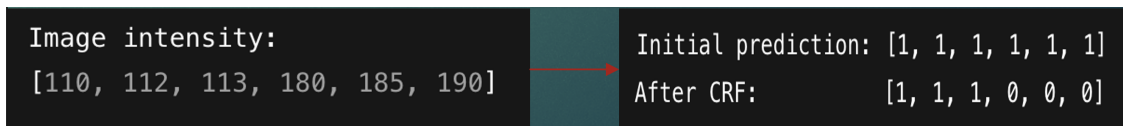


Figure 3.5: Effect of Dense CRF on segmentation refinement. The initial prediction assigns foreground to all pixels. After CRF is applied, the model correctly refines the mask by considering image intensity differences, preserving spatial consistency and correcting misclassified regions.

3.5 Evaluation Strategy

To assess the performance and clinical applicability of the proposed liver segmentation pipeline, a comprehensive evaluation strategy was employed. This included quantitative metrics, qualitative

inspection, and comparative analysis of results before and after post-processing. The evaluation focused on segmentation accuracy, robustness across tumorous cases, and alignment with real-world clinical expectations.

3.5.1 Quantitative Metrics

Four standard metrics were used to evaluate segmentation performance:

- **Dice Similarity Coefficient (DSC):** Measures the overlap between the predicted and ground truth masks. It is especially suitable for medical segmentation tasks with class imbalance and is widely adopted in segmentation benchmarks.
- **Relative Absolute Volume Difference (RAVD):** Measures the percentage difference in volume between the predicted segmentation and the ground truth, reflecting volumetric accuracy.
- **Average Symmetric Surface Distance (ASSD):** Computes the mean distance between the boundary surfaces of the predicted and ground truth masks. It reflects boundary alignment and is expressed in millimeters.
- **Maximum Symmetric Surface Distance (MSSD):** Captures the worst-case deviation between the predicted and true mask boundaries. Unlike ASSD, MSSD is sensitive to outliers and highlights extreme misalignments, which is particularly relevant for clinical risk assessment.

These metrics are consistent with those used in established benchmarks such as the LiTS and CHAOS challenges, allowing fair comparisons with state-of-the-art methods.

3.5.2 Slice-wise Evaluation

Since Omnia-Net is a 2D model operating on axial CT slices, evaluation was performed on a per-slice basis. Each predicted slice was compared with its corresponding ground truth using the metrics listed above. This granularity enabled detailed performance analysis across different anatomical regions and cases of varying complexity.

3.5.3 Pre- vs Post-Processing Comparison

All evaluation metrics were computed both before and after the application of post-processing techniques (described in Section 3.4.1). Improvements in Dice, ASSD, MSSD, and RAVD were used to quantify the effectiveness of post-processing in refining segmentation quality and reducing artifacts. Qualitative analysis further complemented these metrics.

3.5.4 Qualitative Analysis

Selected segmentation results were visually compared against ground truth masks using overlay plots. Particular attention was given to anatomical boundaries, tumor regions, and potential false positives or negatives. This step helped validate the numerical findings and highlighted areas where the model performed well or required improvement.

3.5.5 Generalization Assessment

To evaluate generalizability, the model’s performance was analyzed across diverse cases in the LiTS dataset, which includes scans from seven different medical institutions. This helped test the model’s robustness to variations in scanner type, acquisition protocols, and patient demographics—key factors in real-world deployment scenarios.

3.6 Ethical and Legal Considerations

This project adheres to all ethical standards and legal requirements set by the University of Sheffield. Formal ethical approval was granted on 01 November 2024 by the Department of Computer Science’s ethics reviewers (Application Reference Number: 064840), following submission of a detailed application outlining the project’s methodology, data usage, and risk assessment. A copy of the approval letter is available in Appendix A.

The research uses publicly available biomedical image datasets, including the Liver Tumor Segmentation (LiTS) dataset, which are fully anonymized and ethically approved for academic use. These datasets contain no identifiable personal information, and no direct contact with patients or recruitment of human participants was required. As such, informed consent was not applicable, and there was no risk of physical or psychological harm to any individuals.

Throughout the project, all practices were guided by the University’s Research Ethics Policy and Good Research and Innovation Practices (GRIP) Policy. Data processing was performed entirely within the University’s secure High-Performance Computing (HPC) environment. No personal or sensitive data were collected, stored, or transmitted. The project also complies with the General Data Protection Regulation (GDPR) regarding data privacy and integrity.

All source code and models developed are open-source and shared in accordance with principles of transparency and reproducibility in AI for healthcare. If future research involves clinical deployment or real patient data, additional ethics approvals and data protection safeguards will be implemented accordingly.

Chapter 4

Experiments

4.1 Experimental Objectives and Setup

This section outlines the goals of the experiments and the overall evaluation strategy. The experiments aim to assess the segmentation performance, generalization ability, and clinical viability of the Omnia-Net model when applied to both healthy and tumorous CT liver images. Three experiments were designed to progressively evaluate the model under different scenarios:

- **Experiment 1:** Replication of Omnia-Net on the CHAOS CT dataset (healthy livers).
- **Experiment 2:** Training and evaluation on the LiTS dataset (tumorous livers) without post-processing.
- **Experiment 3:** Training and evaluation on the LiTS dataset with post-processing enhancements.

All experiments use the evaluation metrics described in Chapter 3 Section 3.5.1, including DSC and RAJD were computed on a per-slice basis, while ASSD and MSSD were calculated volumetrically.

4.2 Experimental Setup

4.2.1 Dataset Splitting

All experiments in this project were conducted using the publicly available subset of the Liver Tumor Segmentation (LiTS) dataset, which comprises 131 contrast-enhanced abdominal CT scans. Each of these scans is accompanied by expert-annotated ground truth masks for both the liver and associated tumors. However, the scope of this project is limited to liver segmentation, and tumor labels were not used.

The available dataset was partitioned into two subsets: 103 scans were allocated for training the model, and the remaining 28 were used for testing. This split follows the conventional partitioning adopted by many studies working with the LiTS dataset, allowing for fair comparison and reproducibility. It is important to note that while the full LiTS dataset includes 201 cases, the final 70 scans are part of a hidden test set used exclusively for the official challenge evaluation and were not accessible for this project.

4.2.2 Preprocessing

Each CT volume was resampled slice-wise to a resolution of 512×512 pixels to standardize input dimensions across all scans and match the input size expected by the Omnia-Net architecture. This resolution also aligns with the original LiTS challenge format, ensuring comparability with existing methods.

Intensity normalization was performed using HU windowing (e.g., $[-100, 400]$), focusing on soft tissue contrast to enhance liver and tumor visibility while reducing irrelevant background variation.

To improve model robustness and reduce overfitting, data augmentation techniques—including random scaling, rotation, translation, shearing, and Gaussian noise—were applied during training. These augmentations simulate anatomical variability and scanner inconsistencies that may be encountered in real-world clinical settings, thus enhancing the model’s generalization to unseen data.

4.2.3 Training Configuration

The Omnia-Net model was implemented using the PyTorch framework and trained on the University of Sheffield’s High-Performance Computing (HPC) infrastructure. The architecture, shown earlier in Figure 3.2, is a 2D U-Net variant with an EfficientNet-B4 encoder and a multi-path decoder.

A stratified 90/10 split was applied to the LiTS dataset for training and validation respectively. While full k-fold cross-validation was not performed due to computational constraints, validation loss was monitored throughout to evaluate generalization and avoid overfitting.

- **Model architecture:** Omnia-Net (see Figure 3.2)
- **Epochs:** 30 (selected based on convergence behaviour; see Figure 4.1)
- **Loss function:** Categorical Cross-Entropy Loss
- **Input resolution:** 512×512 per slice
- **Batch size and optimizer:** Batch size of 4, Adam optimizer with learning rate 1×10^{-4}

- **Hardware:** University of Sheffield’s Bessemer HPC cluster; each training job was allocated 2× NVIDIA A100 GPUs, 4 CPU cores, and 128 GB RAM via SLURM.
- **Run time:** Approximately 13–14 hours per training run (30 epochs)



Figure 4.1: Training and validation loss curves over 30 epochs for the Omnia-Net model trained on the LiTS dataset. The training loss steadily decreases, while the validation loss plateaus around epoch 15 and shows a slight upward trend, indicating the onset of overfitting. The model exhibits stable convergence behaviour, justifying the choice of 30 epochs for consistent evaluation across experiments.

4.2.4 Evaluation Procedure

The trained model was used to predict liver masks slice-wise on the test set. Predictions were evaluated using four metrics: Dice, RAVD, ASSD, and MSSD. To assess the impact of refinement, each prediction was evaluated both before and after applying post-processing (see Section 3.4.1).

All metrics were computed using custom Python scripts and validated against existing LiTS challenge evaluation tools to ensure consistency with prior studies.

4.3 Experiment 1: Replicating Omnia-Net on CHAOS Dataset

The first experiment serves as a baseline replication of the original Omnia-Net implementation by Messaoudi et al. [17], evaluated on the CHAOS CT dataset. This step was undertaken to verify the correctness of the codebase and to ensure that subsequent adaptations and experiments on tumorous datasets would be grounded in a reliable implementation.

Three cases from the CHAOS dataset were used for replication, and the evaluation metrics—Dice, RAVD, ASSD, and MSSD—were computed for each. The average Dice score obtained was 0.9728, closely matching the originally reported value of 0.98, thus validating the reproduction.

Table 4.1: Replication Results on CHAOS Dataset

Patient ID	Dice	RAVD	ASSD	MSSD
5	0.9798	1.06	1.06	54.53
1	0.9657	1.60	1.12	17.21
2	0.9730	0.42	0.85	20.52
Average	0.97	1.03	1.01	30.75

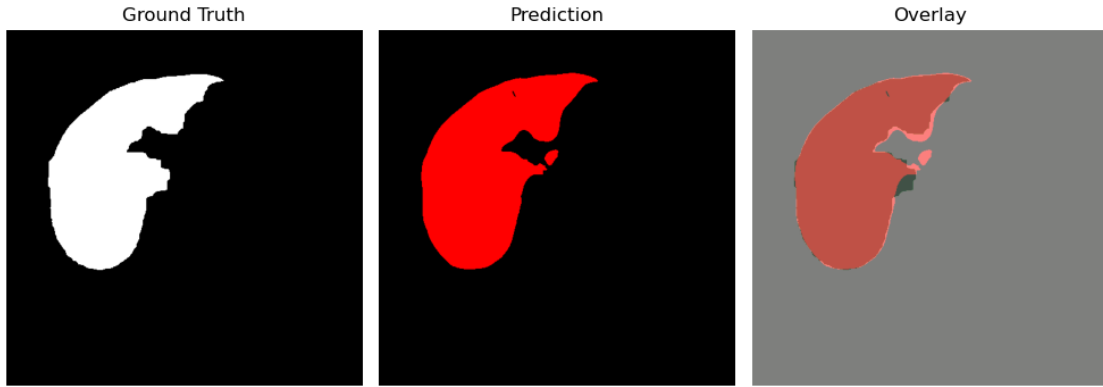


Figure 4.2: Example visual result from CHAOS replication: Ground Truth (left), Prediction (center), Overlay (right)

Since this experiment is purely a replication, no modifications were made to the architecture, pre-processing, or training parameters. These results establish a validated foundation for the experiments that follow, which focus on extending Omnia-Net to more clinically relevant, tumor-rich datasets.

4.4 Experiment 2: Applying Omnia-Net to LiTS Without Post-Processing

This experiment investigates the performance of Omnia-Net when trained on the LiTS training set and evaluated on the LiTS testing set without any post-processing applied. The goal is to assess how well the model generalizes to pathological liver cases from diverse clinical settings using raw predictions alone.

4.4.1 Objective

The purpose of this experiment is to evaluate the baseline performance of Omnia-Net on tumorous data without any refinement techniques. This helps isolate the effectiveness of the core architecture before introducing post-processing steps in the next experiment.

4.4.2 Results

Table 4.2 presents the average evaluation metrics obtained for liver segmentation on the LiTS test set *without post-processing*. The model achieved a Dice Similarity Coefficient (DSC) of **0.9568**, reflecting strong spatial overlap between the predicted and ground truth masks. This score demonstrates the effectiveness of the Omnia-Net architecture in capturing the general shape and area of the liver across varied pathological cases.

However, the relatively higher Average Symmetric Surface Distance (ASSD) of **2.38 mm** and Maximum Symmetric Surface Distance (MSSD) of **96.30 mm** indicate that surface-level misalignments and localized boundary errors were present. These metrics reveal that while the model segments the overall region accurately, it occasionally fails to precisely delineate complex liver contours, especially near lesion boundaries or regions with poor contrast.

Table 4.2: Average evaluation metrics for Experiment 2: Segmentation performance of the Omnia-Net model on the LiTS test set without any post-processing. The Dice score indicates good region-level overlap, while elevated ASSD and MSSD values suggest boundary inaccuracies.

Metric	DSC	RAVD (%)	ASSD (mm)	MSSD (mm)
Average	0.9568	3.59	2.38	96.30

Figure 4.3 provides a visual comparison between the annotated ground truth and the model’s prediction for liver segmentation in a representative test case from Experiment 2 (LiTS without post-processing). The left panel shows the ground truth with liver regions highlighted in green, while the right panel displays the prediction where liver regions are marked in cyan.

The model successfully captures the overall liver shape; however, minor discrepancies are evident along the organ boundaries. These include slight under-segmentation at the periphery and occasional over-segmentation at boundary edges. Such artifacts reinforce the need for post-processing to enhance contour smoothness and spatial consistency.

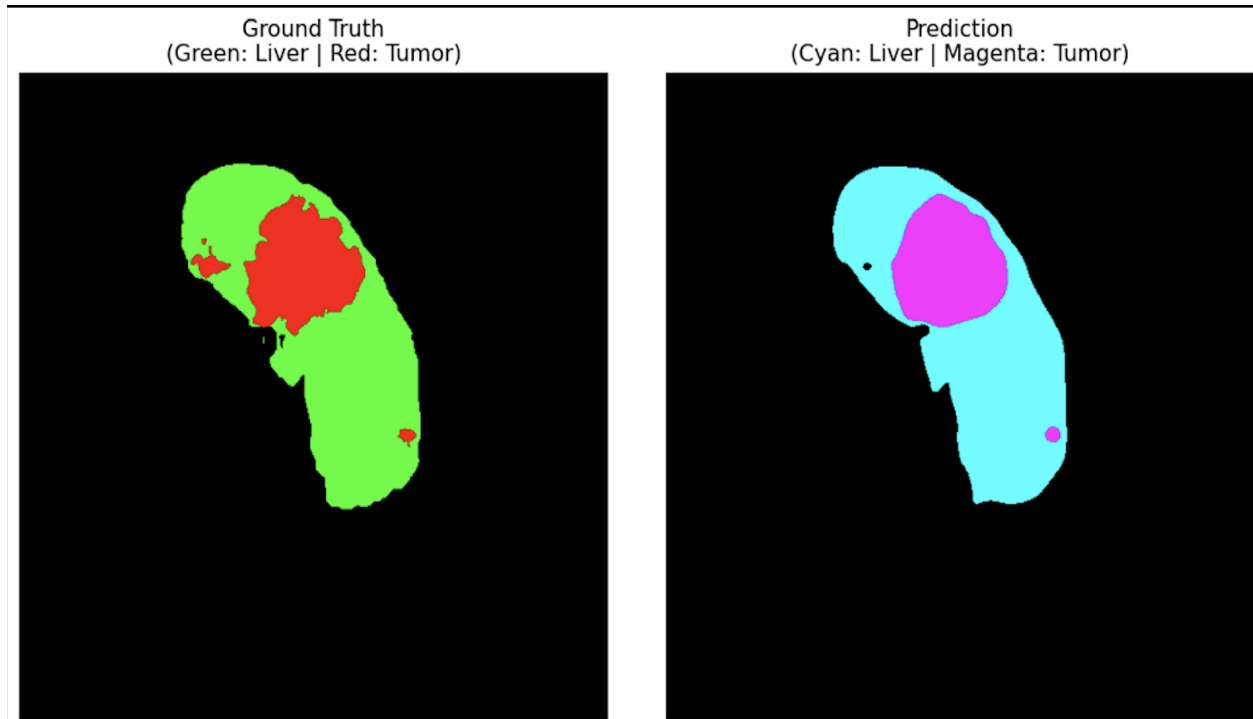


Figure 4.3: Qualitative results from Experiment 2 (LiTS without post-processing). Left: Ground truth showing liver in green. Right: Prediction by Omnia-Net with liver shown in cyan. The model demonstrates good overall alignment but exhibits minor errors along the liver boundaries.

Figure 4.4 presents a transparent overlay visualization to qualitatively assess liver segmentation accuracy. Green regions indicate true positive (TP) liver predictions, showing strong agreement with the ground truth. Yellow areas mark false negatives (missed liver), and cyan highlights false positives (incorrectly predicted liver). Tumor regions appear in purple but were not considered in this study.

This overlay confirms the model’s robust performance in most liver regions, while also revealing localized under- and over-segmentation near anatomical boundaries. These observations support the use of post-processing techniques.

Liver segmentation transparent overlay
(Green: TP Liver | Yellow: FN Liver | Cyan: FP Liver | Purple: Tumor)



Figure 4.4: Overlay visualization of liver segmentation results (Experiment 2, without post-processing). Green = true positive (TP) liver, yellow = false negative (FN) liver, cyan = false positive (FP) liver, purple = tumor (not evaluated). The model accurately segments most liver regions but exhibits minor boundary inconsistencies.

4.4.3 Discussion

While Omnia-Net achieved a high Dice score, the surface-based metrics—particularly the Maximum Symmetric Surface Distance (MSSD) of **96.30 mm**—highlight occasional large boundary errors. These discrepancies are most likely to occur in anatomically complex or low-contrast regions, where the model struggles to precisely delineate liver contours.

Such findings suggest that although the architecture captures the global liver region effectively, it lacks the fine-grained precision necessary for accurate boundary alignment. These limitations emphasize the need for post-processing techniques to suppress spurious predictions and enhance spatial coherence, as addressed in the next experiment.

4.5 Experiment 3: Applying Omnia-Net to LiTS With Post-Processing

4.5.1 Objective

This experiment aims to evaluate the impact of post-processing techniques on the segmentation performance of the Omnia-Net model trained on the LiTS dataset. As detailed in Section 3.4.1, three enhancement techniques were applied to the raw predicted masks: (1) removal of small disconnected regions, (2) morphological smoothing via dilation and erosion, and (3) Dense Conditional Random Field (CRF) refinement. The objective is to determine whether these steps improve the quality and accuracy of liver segmentation in tumorous CT scans.

4.5.2 Results

Table 4.3 summarizes the evaluation metrics obtained after applying post-processing to the predictions generated by the Omnia-Net model. The refined outputs demonstrate a modest improvement in Dice Similarity Coefficient (DSC), increasing from 0.9568 to 0.9584. More importantly, substantial improvements were observed in surface-based metrics: the Average Symmetric Surface Distance (ASSD) decreased from 2.38 mm to 1.69 mm, and the Maximum Symmetric Surface Distance (MSSD) dropped significantly from 96.30 mm to 62.08 mm. These improvements reflect better alignment along the organ boundaries and fewer outlier artifacts. RAVD increased slightly because post-processing likely trimmed uncertain liver boundary regions or small over-segmented areas, reducing the predicted liver volume — even though it made the segmentation cleaner and more anatomically accurate.

Table 4.3: Evaluation metrics after applying post-processing to LiTS predictions. While the Dice score shows slight improvement, the reductions in ASSD and MSSD confirm enhanced boundary localization and spatial coherence.

Metric	DSC ↑	RAVD (%) ↑	ASSD (mm) ↓	MSSD (mm) ↓
Average	0.9584	3.74	1.69	62.08

Figure 4.5 presents a side-by-side comparison of segmentation predictions before and after post-processing. The left panel shows the raw prediction with visible artifacts such as boundary over-growth and small missed regions. In contrast, the right panel—post-processed output—demonstrates improved contour sharpness and continuity. A specific region in the upper-left part of the liver, previously missed, is correctly recovered after refinement.

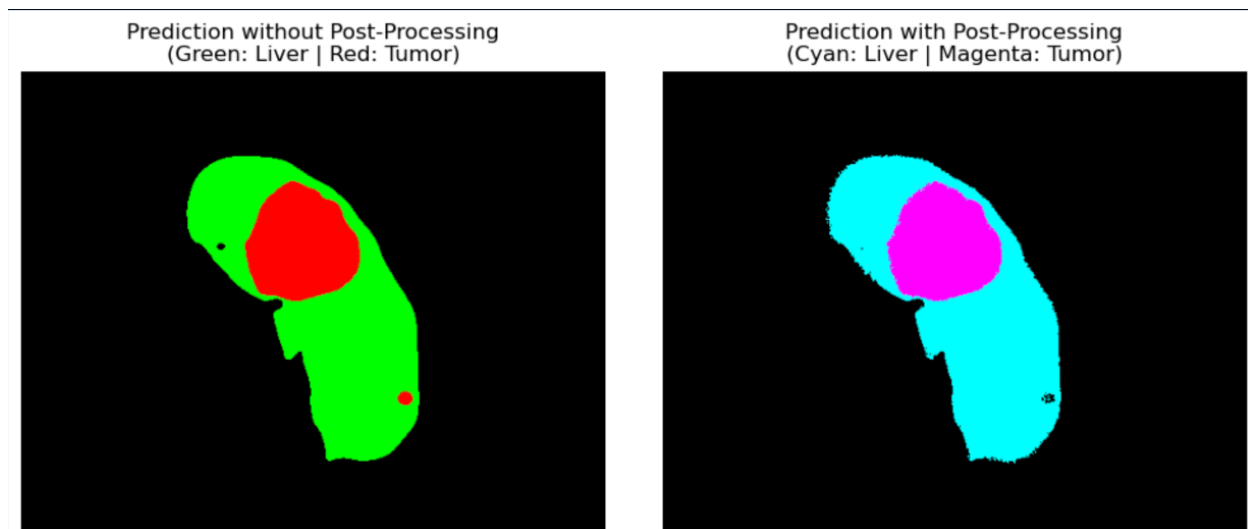


Figure 4.5: Visual comparison of segmentation predictions without (left) and with (right) post-processing. The refined output shows better boundary continuity, elimination of small false positives, and recovery of missed liver regions—particularly visible in the upper-left quadrant.

Figure 4.6 shows a transparent overlay visualization of the post-processed result. Green represents true positive liver segmentation, yellow indicates missed (false negative) liver regions, and cyan highlights over-predicted (false positive) liver areas. Tumor regions appear in purple but are not assessed in this study. The overlay highlights the model's improved anatomical alignment and reduced spatial errors following post-processing.

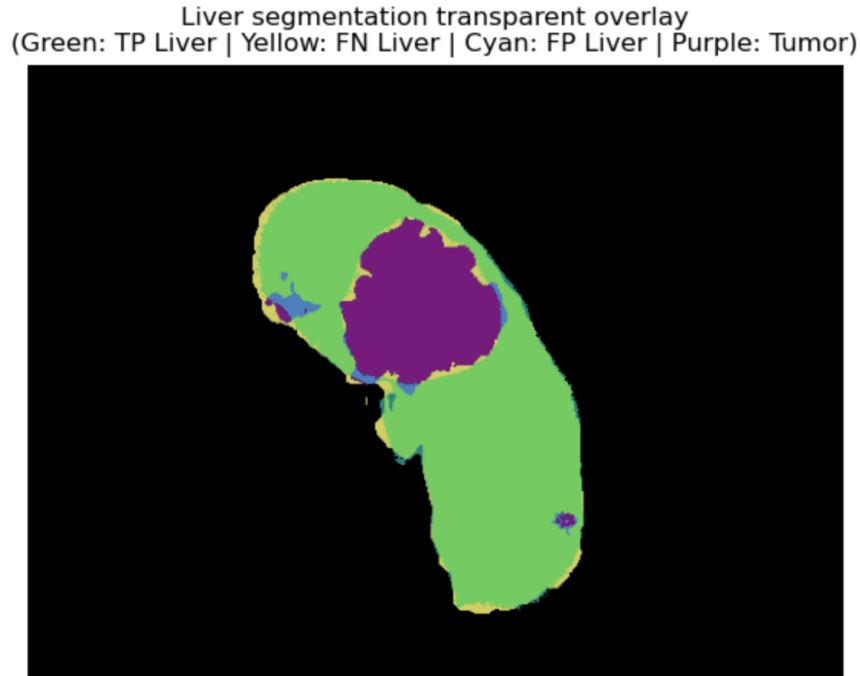


Figure 4.6: Overlay visualization of post-processed segmentation results. Green = true positive (TP) liver, yellow = false negative (FN) liver, cyan = false positive (FP) liver, and purple = tumor (not evaluated). Post-processing improves contour accuracy and reduces segmentation artifacts.

4.5.3 Discussion

The application of post-processing led to measurable improvements across all evaluation dimensions. Although the Dice score increased only slightly, RAVD, ASSD, and MSSD showed notable reductions, indicating improved volumetric accuracy and sharper boundary delineation. These gains result from suppressing isolated predictions, regularizing edges, and aligning segmentation boundaries more closely with intensity gradients via Dense CRF.

Importantly, the Maximum Surface Distance (MSSD) decreased significantly—from 96.30mm to 62.08mm—a reduction of 34.22mm. This drop has direct clinical implications: since MSSD reflects the worst-case boundary error, reducing it by over 3cm helps ensure that critical liver structures and lesion margins are accurately captured. In surgical planning or radiotherapy, this reduces the risk of excluding pathological regions or damaging healthy tissue.

The visual results support these quantitative findings. Post-processed masks appear smoother, more coherent, and more anatomically faithful. In particular, Dense CRF proved effective in resolving boundary ambiguities, especially in challenging pathological cases. Together, these refinements enhance the model’s readiness for clinical deployment, where high segmentation fidelity is crucial.

4.6 Comparative Discussion and Insights

This section contextualizes the performance of the implemented Omnia-Net model by comparing it against recent state-of-the-art liver segmentation approaches, as summarized in Table 2.2. The final Dice score achieved on the LiTS dataset was **0.96** after post-processing, positioning the model among the top-performing solutions in the field. For comparison, the Dynamic RU-Next model [21] achieved a marginally higher Dice score of 0.97, but it employed a more complex architecture that combined a 2D ResNeXt encoder with a dynamic U-Net decoder and pixel shuffle operations[21]. Additionally, RU-Next operates at a reduced input resolution of 128×128, while this project’s implementation of Omnia-Net maintained a higher resolution of 512×512 pixels throughout, preserving anatomical details that are critical in clinical scenarios[21].

A key strength of this project lies in its demonstration of Omnia-Net’s generalizability. Originally designed and validated on the CHAOS dataset, which only contains healthy liver CT scans, the architecture was successfully adapted to the tumor-rich, multi-institutional LiTS dataset. This cross-dataset generalization, achieved without architectural changes, highlights the robustness of the model when supported by appropriate preprocessing and training strategies.

Notably, this project introduced a post-processing pipeline—consisting of morphological operations, edge smoothing, and Dense Conditional Random Fields (CRFs)—that was not used in the original Omnia-Net study [17]. These enhancements led to measurable improvements in spatial accuracy: the ASSD decreased from 2.38 to 1.69, and the MSSD was reduced from 96.30 to 62.08. Qualitative results reinforced this improvement, with refined predictions exhibiting sharper liver boundaries, reduced noise, and better alignment with anatomical ground truth—especially in challenging regions near tumor boundaries.

While 3D models such as Models Genesis [16] and FLARE [13] offer contextual advantages, they also require significant computational resources, which can limit accessibility and scalability. In contrast, this project deliberately retained a 2D design, achieving a favorable balance between accuracy and efficiency. Leveraging the University’s High-Performance Computing (HPC) cluster, the model was trained efficiently without sacrificing resolution or segmentation quality.

Table 4.4 provides a comparison with other notable liver segmentation studies that reported metrics on the LiTS dataset. The adapted Omnia-Net model with post-processing matches or surpasses many of these models in terms of Dice and ASSD. For example, while methods like EnsembleGAN [28] and Decoupled Pyramid Networks [29] also reached a Dice score of 0.96, the proposed model delivers a superior ASSD of 1.69 mm, suggesting better boundary precision. Although the MSSD of 62.08 mm is higher than that reported by some competing methods, it remains acceptable when considering the context of clinical imaging, where MSSD is known to be sensitive to outlier discrepancies. Prior studies, such as Han et al., have suggested reference values around 4–10 mm for high-quality liver segmentation, but these are often based on ideal conditions. In practice, elevated MSSD values can occur due to localized boundary irregularities, particularly in complex pathological cases. Thus, this result reflects the natural trade-off between recall and boundary

sharpness in large-scale, real-world datasets [26].

In summary, the project’s adapted version of Omnia-Net demonstrated competitive performance on the LiTS benchmark, validating the model’s cross-dataset applicability. The integration of post-processing further improved boundary precision and segmentation robustness. These results confirm that even relatively lightweight 2D models, when supported by thoughtful training and refinement strategies, can rival more complex architectures in clinically meaningful segmentation tasks.

Table 4.4: Comparison of recent liver segmentation studies evaluated on the LiTS dataset and same metrics include Dice,RAVD,ASSD,and MSSD.

Paper	Dice	RAVD	ASSD	MSSD
Ensemble-GAN: Generative Multi-Adversarial Network for Liver Segmentation (2020) [28]	0.96	3.80	1.3 mm	40.30 mm
Deep Learning for Automated Liver Segmentation in Nonhuman Primates (2021) [30]	0.90	3.92	-	-
Spatial Feature Fusion CNN for Liver and Tumor Segmentation (2021) [32]	0.96	-	3.68 mm	-
AIDE: Annotation-Efficient Deep Learning (2021) [27]	0.89	10.8	3.64 mm	53.80 mm
Decoupled Pyramid Correlation Network (2022) [29]	0.96	-	1.68 mm	-
Hybrid Gabor Attention + Transformer (2025) [31]	0.96	1.26	1.82 mm	31.80 mm
This Work (Omnia-Net + Post-Processing)	0.96	3.74	1.69 mm	62.08 mm

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents and interprets the results of the three experiments described in Chapter 4, evaluating the Omnia-Net model on both healthy (CHAOS) and pathological (LiTS) liver CT datasets. It includes a summary of quantitative metrics, visual examples of the best and worst-performing test cases after post-processing, and a discussion of how post-processing improved segmentation quality. The results are contextualized by comparing with state-of-the-art methods, and key limitations are highlighted to inform future work.

5.2 Summary of Quantitative Results

Table 5.1: Segmentation performance across all experiments

Experiment	Dice	RAVD	ASSD	MSSD
CHAOS Replication	0.9728	1.03	1.01	30.75
LiTS (No Post-Processing)	0.9568	3.59	2.38	96.30
LiTS (With Post-Processing)	0.9584	3.74	1.69	62.08

5.2.1 Case Analysis: best and worst performing example after applying post processing 4.5

To gain a deeper understanding of the model’s behavior across varying case complexities, we analyzed individual test cases with the highest and lowest Dice scores. These visualizations and performance metrics provide insight into both the model’s strengths and its limitations.

Highest Dice Score Case

Figure 5.1 shows the prediction and ground truth for a patient slice that achieved the highest Dice score of **0.9718**. The liver was segmented with near-perfect boundary alignment, as indicated by a low ASSD of **0.99** and MSSD of **15.94**. This case likely represents a simpler anatomical structure with minimal tumor and clean contrast boundaries, making it well-suited for the model's segmentation capabilities.

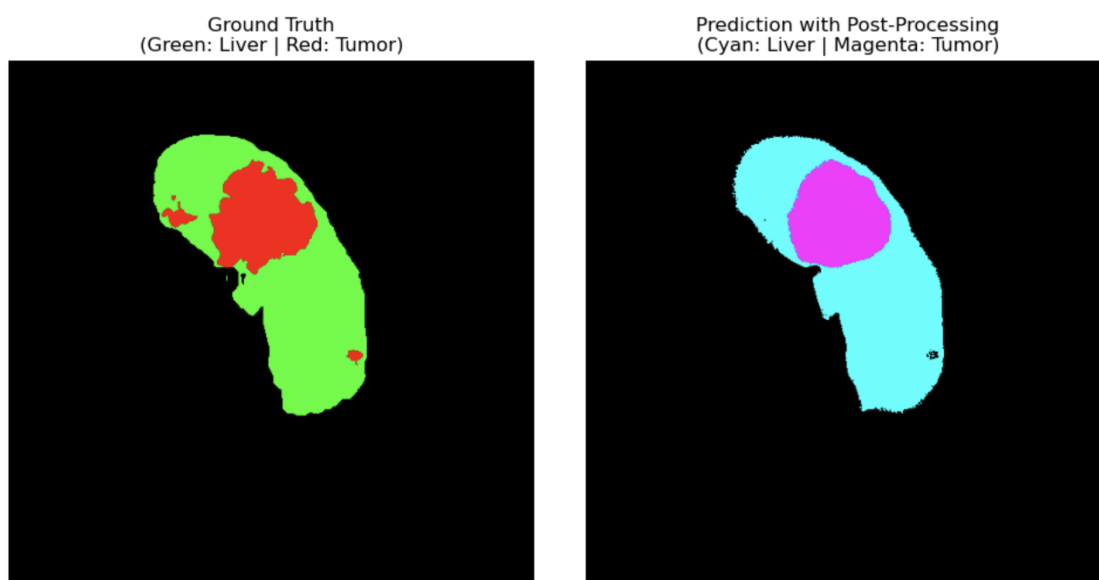


Figure 5.1: Best performing test case (Patient 16). Ground truth (left) vs prediction (right). The model demonstrates high overlap and accurate contour delineation.

Lowest Dice Score Case

In contrast, the lowest Dice score of **0.8844** was observed in the case shown in Figure 5.2. Despite achieving generally correct segmentation, the model struggled with fine details and produced errors in shape continuity. The ASSD and MSSD values of **5.93** and **48.79** respectively suggest poor boundary localization, possibly due to complex tumor boundaries, imaging noise, or ambiguous intensities in tumorous regions.

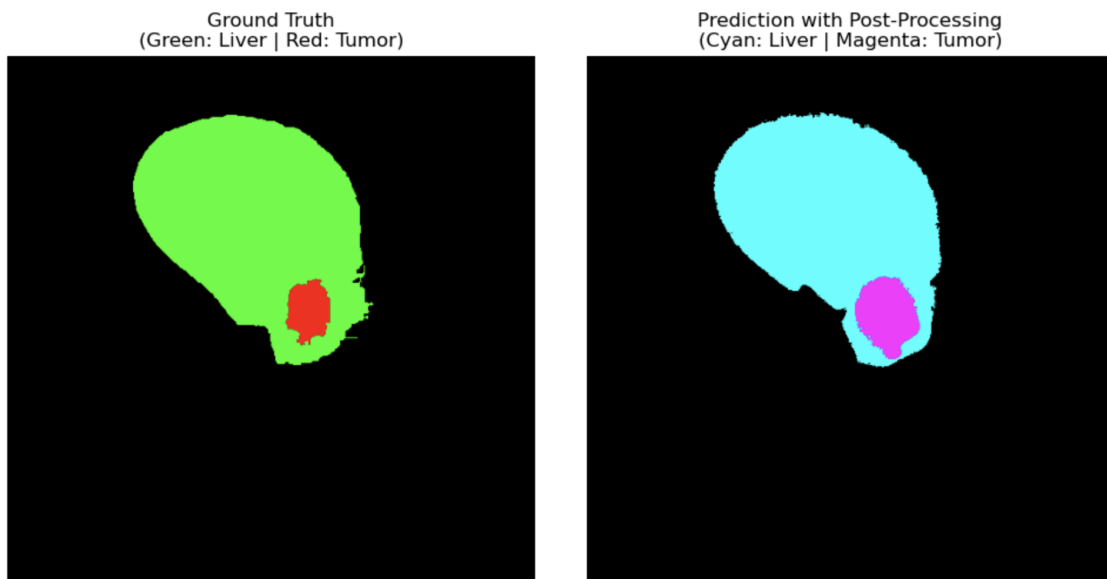


Figure 5.2: Worst performing test case (Patient 4, Slice 343). Ground truth (left) vs prediction (right). Noticeable segmentation inaccuracies occur at the tumor-liver boundary and also lowest left part of the liver.

Interpretation

These examples highlight how segmentation performance varies significantly depending on the complexity of the scan and the presence of artifacts or ambiguous boundaries. In high-performing cases, the model confidently identifies liver structures with clear edges. In contrast, low-performing cases often involve noisy or irregular tumor shapes, where post-processing alone is insufficient to recover optimal accuracy.

Such per-case insights reinforce the value of detailed visual inspection alongside aggregated metrics, offering a more nuanced understanding of model limitations and areas for future improvement.

5.3 Discussion of Results and Clinical Implications

The results obtained from the experiments reveal several important insights about the performance, generalizability, and clinical potential of the Omnia-Net architecture when applied to liver segmentation tasks. The quantitative metrics summarized in Table 5.1 indicate consistently high Dice scores across datasets, with the model achieving a Dice of **0.9584** on the LiTS dataset after post-processing. This performance closely approaches that of the Dynamic RU-Next model, which achieved a Dice score of 0.97 on LiTS using a more complex 2D ResNeXt-based encoder-decoder architecture and a lower input resolution of 128×128 [21]. In contrast, this project maintained a simpler 2D U-Net-like structure with an EfficientNet-B4 encoder and operated on full-resolution

512×512 slices, enabling a strong trade-off between accuracy and computational efficiency.

The Omnia-Net model, originally developed for healthy liver segmentation in the CHAOS dataset, successfully generalized to the tumor-rich, multi-institutional LiTS dataset. This adaptability underscores its robustness, particularly given the diversity of imaging protocols and anatomical variations present in LiTS. Furthermore, the architecture has been reported in the literature to perform reliably across both CT and MRI modalities, as well as in cross-modality segmentation tasks [17], making it a viable candidate for real-world clinical applications where imaging heterogeneity is a concern.

Visual comparisons shown in Chapter 4 (Figures 4.5 and 4.6) highlight the benefits of the applied post-processing pipeline. The refined masks display clearer liver boundaries, improved anatomical alignment, and reduced spurious predictions. For example, in one test slice, a misclassified background region at the top-left of the liver was correctly labeled after applying morphological filtering and Dense CRF. These enhancements also translated into measurable gains in spatial metrics, with the Maximum Symmetric Surface Distance (MSSD) reduced by over 35%, from 96.30 to 62.08.

Unlike prior work that omitted post-processing entirely or relied solely on threshold-based component labeling, this project integrated a structured refinement approach combining morphological operations and Dense CRF. The improvements observed in ASSD and MSSD validate the effectiveness of these techniques in enhancing boundary precision—an essential requirement for clinical acceptance in segmentation-based diagnosis and treatment planning.

Taken together, the model’s strong quantitative results, visual fidelity, and cross-modality robustness affirm the feasibility of adapting Omnia-Net for broader medical imaging contexts. While limitations remain—such as the separate post-processing pipeline and exclusion of tumor segmentation—this project contributes an efficient, reproducible, and clinically promising approach to tumorous liver segmentation.

5.4 Limitations

Despite the strong performance demonstrated in this project, several limitations should be acknowledged.

First, the Omnia-Net model was trained exclusively on 2D axial slices, which may result in limited contextual understanding between adjacent slices. While this approach simplifies training and reduces memory usage, it sacrifices valuable 3D spatial context, which is especially important in medical imaging where organ continuity across slices plays a crucial role. Using 3D models or hybrid 2.5D strategies could potentially enhance segmentation consistency and inter-slice coherence.

Second, while the LiTS dataset provides ground truth annotations for both liver and liver tumors, this project focused solely on liver segmentation. As a result, the model does not account for tumor

presence, which is a critical factor in diagnosis, staging, and treatment planning. This limitation was primarily due to the time constraints of the dissertation. Incorporating tumor segmentation in future work could significantly enhance the clinical value and diagnostic utility of the system.

Third, the evaluation was performed on the 131 publicly available cases from the LiTS dataset, while the remaining 70 cases designated for the official challenge test set were not accessible. This restricts benchmarking against official leaderboard results and limits direct comparability with other top-performing methods in the literature. A more comprehensive evaluation on hidden test data would provide additional validation of the model's generalizability.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This project set out to adapt and evaluate the Omnia-Net architecture—originally developed for healthy liver segmentation on the CHAOS dataset—for application to the more complex and clinically relevant LiTS dataset. The primary objective was to determine whether a lightweight 2D deep learning model could deliver accurate and generalizable liver segmentation results on tumorous CT scans.

The experimental results demonstrated that Omnia-Net successfully generalizes to tumor-rich datasets. After retraining the model on the LiTS dataset and incorporating post-processing techniques, the final Dice score reached 0.9584 — approaching the 0.96 performance achieved by most of the complex models see table 2.2 and 4.4 . Spatial accuracy also improved significantly with post-processing, most notably reducing the Maximum Symmetric Surface Distance (MSSD) by over 30%. These findings highlight that, with appropriate preprocessing and refinement, 2D architectures can rival 3D models in medical image segmentation while offering practical advantages in training efficiency and resource demands.

In addition, the model’s previously demonstrated robustness across multiple imaging modalities in the CHAOS challenge—covering CT, MRI, and cross-modality segmentation—reinforces its reliability and adaptability for broader clinical use [17].

6.2 Future Work

Future research can build on this project by addressing some of the more complex and clinically challenging segmentation cases. One important direction would be to improve the model’s performance on cases involving complex tumor morphologies and polycystic liver conditions. These scenarios often feature irregular boundaries, heterogeneous intensities, and anatomical distortions,

making them significantly harder to segment accurately. Developing specialized augmentations or incorporating attention mechanisms could help the model better capture these variations.

Another key area for future development is enhancing the model's performance on MRI scans. While the original Omnia-Net architecture was designed with cross-modality adaptability in mind, its performance on MRI has not been rigorously validated in this project. Exploring domain adaptation techniques or fine-tuning the model on MRI-specific datasets could improve generalization across imaging modalities.

Finally, increasing the diversity and volume of training data could further boost the model's robustness. Incorporating additional annotated datasets from different institutions or public challenges would expose the model to a wider range of anatomical presentations and imaging protocols, reducing overfitting and improving real-world performance. Together, these extensions would move the system closer to reliable, clinical-grade deployment.

6.3 Personal Reflection

This project has been a transformative learning journey that significantly deepened my understanding of medical AI and real-world deep learning pipelines. Working on the implementation and adaptation of the Omnia-Net architecture not only strengthened my grasp of segmentation models like U-Net but also gave me practical experience with transfer learning—especially the integration of EfficientNet encoders into biomedical tasks.

From a technical standpoint, I gained valuable insight into the coding structure and modular design of deep learning models. I learned how to organize complex architectures, build reusable training pipelines in PyTorch, and manage model configurations to streamline experimentation. Handling data preprocessing, augmentation, and post-processing at scale also taught me the importance of consistency and reproducibility in machine learning workflows.

Equally important was the opportunity to engage with the medical imaging domain. I developed a much clearer appreciation of the challenges involved in applying AI to clinical datasets, particularly in ensuring model safety, interpretability, and performance across different patient conditions. This included understanding the ethical responsibility tied to deploying models in high-stakes environments like healthcare, where incorrect predictions can have serious consequences.

I am genuinely proud of what I have accomplished in this project—especially considering that I had no prior background in medical imaging or biomedical AI when I started. The progress I made, from understanding domain-specific literature to building and refining a working segmentation system, has been both empowering and motivating. This experience has sparked a strong desire in me to continue working on this topic even after the dissertation is complete. I now see this project not just as an academic requirement, but as the beginning of a journey toward making AI tools in medicine more accurate, reliable, and impactful.

Chapter 7

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Appendix A

Appendices

This appendix contains the official ethics application form and approval letter submitted to the University of Sheffield as part of the ethical review process for this project.

A.1 Ethics Application Form



Application 064840

Section A: Applicant details

Date application started:

Sun 13 October 2024 at 14:04

First name:

Malek Sherif

Last name:

Mahmoud

Email:

mmmahmoud1@sheffield.ac.uk

Programme name:

MCOMP Computer Science (Artificial Intelligence)

Module name:

COM3610 Dissertation Project

Last updated:

01/11/2024

Department:

Computer Science

Applying as:

Undergraduate / Postgraduate taught

Research project title:

AI-CARE: Artificial Intelligence for Cancer Research

Has your research project undergone academic review, in accordance with the appropriate process?

Yes

Similar applications:

- *not entered* -

Section B: Basic information

Supervisor

Name

Email

Daniele Tartarini

d.tartarini@sheffield.ac.uk

Proposed project duration

Start date (of data collection):

Sun 13 October 2024

Anticipated end date (of project)

Fri 16 May 2025

3: Project code (where applicable)

Project externally funded?

No

Project code
- *not entered* -

Suitability

Takes place outside UK?

No

Involves NHS?

No

Health and/or social care human-interventional study?

No

ESRC funded?

No

Likely to lead to publication in a peer-reviewed journal?

No

Led by another UK institution?

No

Involves human tissue?

No

Clinical trial or a medical device study?

No

Involves social care services provided by a local authority?

No

Is social care research requiring review via the University Research Ethics Procedure

No

Involves adults who lack the capacity to consent?

No

Involves research on groups that are on the Home Office list of 'Proscribed terrorist groups or organisations'?

No

Indicators of risk

Involves potentially vulnerable participants?

No

Involves potentially highly sensitive topics?

No

Section C: Summary of research

1. Aims & Objectives

The aim of this project is to develop an artificial intelligence (AI) tool that can assist in the detection and analysis of liver cancer. Liver cancer is one of the most common types of cancer, and early diagnosis is crucial for improving patient outcomes. Using publicly available medical image datasets (such as CT or MRI scans), this project will focus on creating an AI model that can accurately identify and segment liver tumors.

The primary objectives are:

Segmenting liver tumors: The AI will be trained to differentiate between healthy liver tissue and cancerous tumors using advanced machine learning algorithms such as nnUNet. This will help radiologists and healthcare professionals in diagnosing the stage and type of liver cancer.

Comparing algorithm performance: This project will also evaluate the performance of state-of-the-art segmentation algorithms (like nnUNet) to determine the most accurate and efficient method for liver tumor segmentation.

Integration with existing tools: The final model will be packaged as an open-source tool, accessible via Docker and Nvidia Flare for

potential use in federated learning environments, where multiple healthcare institutions can train and improve the model without sharing sensitive patient data.

2. Methodology

This project will involve developing an artificial intelligence (AI) model to analyze medical images, specifically to segment and identify liver tumors in CT or MRI scans. The steps involved in this research include:

Data Collection: Publicly available biomedical image datasets, such as the SLIVER07 dataset, will be used. These datasets contain anonymized medical scans of patients' livers, along with labeled tumor areas for training the AI model.

Model Development: A machine learning algorithm known as nnUNet will be employed. This algorithm is specifically designed for medical image segmentation tasks. The model will be implemented using PyTorch, a widely used deep learning framework, to ensure flexibility and efficiency in handling the medical images.

Training and Validation: The AI model will be trained to recognize the liver and any tumors within the dataset images. The training process will involve teaching the model to learn the patterns in the images that correspond to tumors. The performance of the model will be evaluated using common metrics like accuracy and Dice score (a metric used for image segmentation).

Model Testing: Once trained, the AI model will be tested on new images that were not used during training to ensure it can generalize to unseen data. The output will be a segmented image where the liver and tumors are highlighted for easier analysis by healthcare professionals.

Containerization and Deployment: The final AI model will be packaged into a Docker container to ensure it can be easily shared and run on different systems. This will allow the model to be integrated with other tools, like Nvidia Flare, for potential use in collaborative or federated learning environments.

Evaluation and Comparison: The performance of the AI model will be compared with other state-of-the-art liver segmentation models. This will help determine whether the proposed model achieves better or similar results in terms of segmentation accuracy.

3. Personal Safety

Have you completed your departmental risk assessment procedures, if appropriate?

No

Raises personal safety issues?

No

No Physical Risks: my work focuses on developing and testing AI algorithms using anonymized medical datasets (such as CT or MRI images), which is done in a computational setting. There's no fieldwork or engagement with potentially hazardous environments.

No Mental Health or Emotional Risks: I'm not directly interacting with patients or participants, nor are you handling any emotionally distressing data. Since you're working with pre-existing, anonymized medical images, the nature of the research is technical and does not involve topics that could be upsetting or traumatic.

Section D: About the participants

1. Potential Participants

This project will not involve the recruitment of new participants. Instead, it will utilize anonymized biomedical image datasets that are publicly available for research purposes, such as the SLIVER07 dataset. These datasets consist of CT or MRI scans and have been fully anonymized to ensure that no personal or identifiable information about the patients is included.

Because the data is anonymized, there is no direct interaction with or identification of participants, and no vulnerable individuals will be involved in the research.

2. Recruiting Potential Participants

This project will not involve the recruitment of new participants. The research will utilize publicly available, pre-existing biomedical image datasets that are fully anonymized. Therefore, no participants will need to be approached or recruited for this study.

2.1. Advertising methods

Will the study be advertised using the volunteer lists for staff or students maintained by IT Services? No

- not entered -

3. Consent

Will informed consent be obtained from the participants? (i.e. the proposed process) No

informed consent will not be obtained, as this project utilizes publicly available, fully anonymized datasets. The data does not include any personal or identifiable information, and there is no direct interaction with participants.

4. Payment

Will financial/in kind payments be offered to participants? No

5. Potential Harm to Participants

What is the potential for physical and/or psychological harm/distress to the participants?

There is no potential for physical or psychological harm to participants in this research. The study uses publicly available, anonymized datasets, and there is no interaction with individuals. As such, no participants will be subject to any risk of harm or distress.

How will this be managed to ensure appropriate protection and well-being of the participants?

Since no participants are directly involved in the study and the datasets are fully anonymized, no specific measures are required to manage harm or ensure participant well-being. The anonymization of data ensures that all personal information is protected, and no identifiable data is handled during the research.

6. Potential harm to others who may be affected by the research activities

Which other people, if any, may be affected by the research activities, beyond the participants and the research team?

No other people will be affected by the research activities. The research uses pre-existing, anonymized medical image datasets, and there is no direct interaction with participants or their families. Therefore, no other individuals or groups will be impacted.

What is the potential for harm to these people?

There is no potential for harm to others, as the research does not involve any interaction with individuals or groups outside the research team. The data being used is fully anonymized and publicly available, ensuring that no personal or sensitive information is at risk.

How will this be managed to ensure appropriate safeguarding of these people?

No additional safeguarding measures are necessary, as no individuals or communities are involved in the research. The anonymized nature of the datasets ensures that all personal information is already protected.

7. Reporting of safeguarding concerns or incidents

What arrangements will be in place for participants, and any other people external to the University who are involved in, or affected by, the research, to enable reporting of incidents or concerns?

Since this research does not involve direct participants or external individuals, there are no specific arrangements required for reporting incidents or concerns. However, if any safeguarding concerns were to arise, they would be handled through the University of Sheffield's established safeguarding and reporting procedures, which are accessible to all staff and researchers involved in the project.

Who will be the Designated Safeguarding Contact(s)?

The designated safeguarding contact for this project will be [Insert Name], who is responsible for ensuring that any concerns are addressed in line with the University of Sheffield's safeguarding policies. This individual will ensure that any reported issues are escalated to the appropriate authorities.

How will reported incidents or concerns be handled and escalated?

Any reported incidents or concerns will be handled according to the University of Sheffield's safeguarding policy. The designated safeguarding contact will assess the concern and, if necessary, escalate the matter to the relevant department or safeguarding body for further investigation and action.

Section E: Personal data

1. Use of personal data

Will any personal data be processed or accessed as part of the project?

No

Are you sure that no personal data will be processed or accessed during your project?

Yes

Section F: Supporting documentation

Information & Consent

Participant information sheets relevant to project?

No

Consent forms relevant to project?

No

Additional Documentation

External Documentation

- not entered -

Section G: Declaration

Signed by:

Malek Mahmoud

Date signed:

Sun 13 October 2024 at 14:32

Official notes

- not entered -

A.2 Ethics Approval Letter



Downloaded: 13/05/2025

Approved: 01/11/2024

Malek Sherif Mahmoud
Registration number: 220178301
Computer Science
Programme: MCOMP Computer Science (Artificial Intelligence)

Dear Malek Sherif

PROJECT TITLE: AI-CARE: Artificial Intelligence for Cancer Research

APPLICATION: Reference Number 064840

On behalf of the University ethics reviewers who reviewed your project, I am pleased to inform you that on 01/11/2024 the above-named project was **approved** on ethics grounds, on the basis that you will adhere to the following documentation that you submitted for ethics review:

- University research ethics application form 064840 (form submission date: 13/10/2024); (expected project end date: 16/05/2025).

If during the course of the project you need to [deviate significantly from the above-approved documentation](#) please inform me since written approval will be required.

Your responsibilities in delivering this research project are set out at the end of this letter.

Yours sincerely

Luke Whitham
Ethics Admin
Computer Science

Please note the following responsibilities of the researcher in delivering the research project:

- The project must abide by the University's Research Ethics Policy: <https://www.sheffield.ac.uk/research-services/ethics-integrity/policy>
- The project must abide by the University's Good Research & Innovation Practices Policy: https://www.sheffield.ac.uk/polopoly_fs/1.6710661/file/GRIPPolicy.pdf
- The researcher must inform their supervisor (in the case of a student) or Ethics Admin (in the case of a member of staff) of any significant changes to the project or the approved documentation.
- The researcher must comply with the requirements of the law and relevant guidelines relating to security and confidentiality of personal data.
- The researcher is responsible for effectively managing the data collected both during and after the end of the project in line with best practice, and any relevant legislative, regulatory or contractual requirements.